

IMPERIAL COLLEGE LONDON

DYSON SCHOOL OF ENGINEERING

A Strategic Engineering Analysis of Modular Desalination Plant Investments under Demand Uncertainty

Authors:

Arancha Ramirez (CID: 02122361)

Date: June 4, 2026

Abstract

Large modular desalination investments must be made under substantial demand and energy-price uncertainty over multi-decade horizons. Conventional Net Present Value analysis treats these projects as static commitments, underweighting the value of managerial flexibility to defer, expand, or abandon capacity as information accumulates.

This thesis develops and applies an integrated framework for valuing flexibility in modular seawater reverse-osmosis investments. The framework combines three analytical layers — real-options valuation through paired Monte Carlo simulation, robustness assessment through distributional metrics and deterministic sensitivity analysis, and a four-axis multi-criteria decision analysis — and delivers them through an interactive web-based Decision-Support Tool designed for both specialist and non-specialist users. The framework is applied to the Torrevieja desalination plant in southeastern Spain across a 15-year horizon.

The case study produces an Expected Value of Flexibility of approximately €118 million, with the flexible design outperforming the rigid design in 81% of 2,000 Monte Carlo trials and remaining positive across $\pm 20\%$ parameter perturbations. The flexible design leads on the economic, environmental, and operational multi-criteria axes; the rigid design leads on predictability owing to its tighter NPV dispersion. A formative usability evaluation with ten participants and a practitioner interview confirms that the Decision-Support Tool achieves its primary specialist objective and partially achieves the secondary accessibility objective.

The thesis contributes a unified framework integrating three previously-isolated decision-support layers, a calibrated techno-economic model of SWRO capacity expansion, a usability-evaluated decision-support artefact, and an empirical assessment of when modular flexibility is valuable. The framework is generalisable to any infrastructure investment with sequential capacity-expansion options.

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Defining flexibility, robustness and sustainability in this thesis	1
1.3	Research aim and objectives	1
1.4	Scope: Why SWRO Desalination, why Spain, what is out of scope	2
1.5	Contributions of this project	2
2	Literature Review	3
2.1	Flexibility & ROA	3
2.2	Robustness and Decision-making under Uncertainty	3
2.3	Multi-criteria Decision Frameworks	4
2.4	Research Gaps	5
2.5	Research Approach	6
3	Methodology	6
3.1	Framework Overview	6
3.2	Phase 1: Stochastic representation of uncertainty	7
3.3	Phase 2: Real options analysis and flexibility valuation	8
3.4	Phase 3: Robustness assessment	8
3.5	Phase 4: Multi-criteria decision analysis	9
4	Case Study: Expansion of the Torrevieja SWRO Desalination Plant	9
4.1	Plant context and the expansion question	9
4.2	Plant alternatives and cost model	10
4.3	Stochastic calibration: demand and electricity price	11
4.3.1	Water demand	11
4.3.2	Electricity price	11
4.3.3	Active calibration selection	12
4.4	Decision-rule parameters	13
4.5	Financial model and evaluation	14
4.5.1	Revenue	14
4.5.2	Variable cost	15
4.5.3	Capital expenditure	15
4.5.4	Net Present Value (NPV)	15
4.5.5	Flexibility Valuation	15
4.6	Multi-criteria decision analysis: criterion specification	16
5	Decision-Support Tool	17
5.1	Motivation and design objectives	17
5.2	Architecture and technology	17
5.3	Tool description	18
5.3.1	Tab 1: About	18
5.3.2	Tab 2: Worked Example	18
5.3.3	Tab 3: Inputs	18
5.3.4	Tab 4: Uncertainty	18
5.3.5	Tab 5: Decision	19
5.4	Human-factors design choices	19
5.5	Usability evaluation	20

5.5.1	Evaluation methodology	20
5.5.2	Findings	21
5.5.3	Practitioner interview	23
5.6	Summary of findings	23
6	Results & Discussion	24
6.1	NPV and Financial Comparison	24
6.2	Distributional Risk and Per-trial Behaviour	26
6.3	Sensitivity analysis	27
6.3.1	Deterministic parameter sensitivity	27
6.4	Multi-criteria perspective	28
7	Conclusion	29
7.1	Overview and contributions	29
7.2	Study limitations	30
7.3	Further work	31
7.4	Professional reflection	32
7.5	Project Timeline and Workflow	32
8	Acknowledgements	33
9	Appendices	38
9.1	Nomenclature	38
9.2	Stochastic Calibration: Source Data	40
9.2.1	Water demand	40
9.2.2	Electricity price	40
9.3	Decision-Rule and Operational Parameter Sources	41
9.4	Model Verification and Cross-Validation	43
9.5	Decision-Support Tool	43

1 Introduction

1.1 Motivation

In an attempt to mitigate increasing water scarcity in regions where surface and groundwater resources are fully allocated or in decline, Seawater reverse osmosis (SWRO) has become a key water supply technology in water-scarce coastal regions. Global installed capacity exceeded 95 million m³/day in 2022 and continues to grow, with Spain, the Middle East, and North Africa accounting for the largest concentrations [1][2]. Driving this growth are converging structural pressures: unreliable surface and groundwater availability, climate-driven shifts in precipitation and drought frequency [3][4], and population and agricultural demand growth in arid coastal economies.

However, SWRO plants are capital-intensive, energy-intensive, and long-lived. Investment decisions are therefore made on the basis of demand and cost expectations that extend well beyond the horizon over which those variables can be reliably forecast. The result is deep uncertainty: the planner cannot assign reliable probability distributions to future states, and several plausible scenarios must be entertained simultaneously [5][6].

Conventional planning responds poorly to this uncertainty. The dominant ‘predict and provide’ approach selects a single demand forecast, sizes a rigid plant against it, and proceeds. This leads to two possible failures: over-investment when demand grows slower than the forecast, or under-investment when demand or energy-costs exceed the design, forcing emergency procurement at a premium cost [7][8][9]. Both failures reflect the same underlying weakness: the absence of designed-in adaptability.

1.2 Defining flexibility, robustness and sustainability in this thesis

Three terms recur throughout this thesis and require precise definitions.

Flexibility is the capacity of an engineered system to be modified, expanded, deferred, or reconfigured after initial deployment in response to information that was unavailable at the time of the design decision [10][11]. Flexibility is a property of the design, not of operations; it must be deliberately embedded through architectural choices such as modularity, oversized interfaces, or staged commitments.

Robustness is the capacity of a system to maintain acceptable performance across a wide range of plausible futures, without requiring foreknowledge of which future will occur [3][12]. Robustness is evaluated rather than designed: a candidate system is stress-tested against many scenarios and judged on its worst-case or distributional performance. The two properties are complementary, not equivalent. A flexible system can adapt as uncertainty resolves; a robust system performs adequately whether or not adaptation occurs. A well-designed infrastructure asset typically requires both.

Sustainability is taken here in its tri-axis sense — economic, environmental, and social viability over the asset lifecycle [13][14]. It is not treated as a separate methodology but as the evaluation space against which flexible and robust designs are ultimately judged.

1.3 Research aim and objectives

The aim of this project is to develop and apply an integrated framework that combines flexibility valuation, robust uncertainty modelling, and multi-criteria decision analysis to support infrastructure planning decisions under deep uncertainty, and to demonstrate it on a representative water infrastructure case.

Four objectives drive this project:

1. Establish the methodological gap by reviewing Real Options Analysis, robust decision-making, and Multi-Criteria Decision Analysis as applied to water infrastructure, identifying the limits of their current application in isolation.
2. Develop a unified framework that integrates the three methodologies within a single computational protocol.
3. Implement the framework as a calibrated techno-economic model of SWRO desalination capacity expansion, with explicit modular-expansion trigger logic and stochastic demand and energy-price processes.
4. Apply the framework to a representative case to produce an empirical assessment of when modular flexibility is valuable, and deliver a usability-evaluated decision-support tool that exposes the framework’s outputs to specialist and non-specialist users.

1.4 Scope: Why SWRO Desalination, why Spain, what is out of scope

This case study focuses on SWRO desalination, selected because it is capital-intensive, modularly extensible, and operates under coupled demand and energy-price uncertainty — the configuration the framework is designed to address. Other desalination technologies (multi-stage flash, multi-effect distillation, brackish-water RO) share the broader uncertainty structure but differ in cost dynamics and are not modelled here.

The analysis treats the investment decision of a single asset owner over a 15-year horizon. Inter-utility coordination, water-market trading, and political-economy questions of water allocation are out of scope. Demand and electricity price are modelled stochastically; capital costs, discount rate, and operating parameters are treated deterministically with sensitivity analysis. Regulatory, technological, and climate-policy uncertainty enter through scenario regimes rather than continuous stochastic processes — a deliberate simplification discussed in Section 3.

1.5 Contributions of this project

The four objectives above produce four corresponding contributions:

1. A unified framework that combines Real Options Analysis, Monte Carlo robustness assessment, and multi-criteria decision analysis within a single computational protocol, addressing the integration gap identified in the literature.
2. A bottom-up techno-economic model of SWRO desalination capacity expansion, calibrated against published cost and operational data, with explicit modular-expansion triggering logic and stochastic demand and energy-price processes.
3. A decision-support artefact that exposes flexibility value, robustness metrics, and multi-criteria trade-offs to specialist and non-specialist users, validated through a structured ten-participant usability study and a complementary practitioner interview.
4. An empirical assessment of when modular flexibility is valuable in SWRO investment, including the conditions under which a modular design dominates a rigid one and where this conclusion is sensitive to assumptions.

2 Literature Review

2.1 Flexibility & ROA

The concept of flexibility as an economically valuable optionality originated in financial economics. Black (1973) established that the right but not the obligation to take a future action carries quantifiable value when the underlying state variable is uncertain [15]. This was further extended to capital investment decisions by formalising the term ‘Real Options Analysis’ (ROA) and demonstrating that conventional discounted cash-flow analysis systematically undervalues investments where decision discretion can be exercised in response to evolving information.

Water-sector applications have emerged in parallel but with less methodological refinement. Early work by Suttinon and Nasu showed that ROA can increase the value of industrial water infrastructure by allowing utilities to defer, expand or adapt investments in response to economic and regulatory changes, rather than committing to rigid capacity plans. Their results challenge the static assumptions that underpin conventional Net Present Value (NPV) approaches and demonstrate that flexibility can materially alter investment decisions [11].

Marques et al. further illustrate this by showing that embedding an expansion option helps water utilities avoid future capacity shortages and realise long-term cost efficiencies [16]. Building on this, Pachos et al. formalise how ROA can balance the risks of under- and over-capacity, providing a decision framework that accounts for the asymmetric consequences of designing too little or too much infrastructure under deep uncertainty [8].

Zhang and Babovic offer one of the most comprehensive ROA applications in water planning, demonstrating that ROA can evaluate water technology pathways across economic, political, and social dimensions, thereby supporting more adaptive and resilient systems [17]. Similarly, Deng et al. show that ROA enables planners to incorporate learning over time, reducing upfront capital commitments while increasing long-term system value as uncertainty about climate and demand conditions resolves [18].

Most recently, Najafi and Aghaei applied a Least-Squares Monte Carlo (LSMC) ROA framework to a photovoltaic-powered RO plant in Saudi Arabia, valuing the option to defer the investment across three power-supply configurations and demonstrating that the embedded deferral option restored economic viability to two configurations that conventional DCF analysis would have rejected. Najafi and Aghaei’s contribution is the methodological transfer of LSMC into SWRO; however, their case treats demand as deterministic and varies only the power-supply configuration, leaving the demand-side option value uninvestigated — the gap the present work addresses.

ROA is an essential tool for valuing flexibility, but insufficient on its own to guide long-term water planning. Its strengths are maximised when integrated with other methodologies — highlighting the need for a unified, cross-methodological approach.

2.2 Robustness and Decision-making under Uncertainty

Robust decision frameworks identify which design performs acceptably across the widest range of plausible futures, regardless of whether adaptation is feasible or whether the underlying probability distributions are known. While ROA presumes a probability distribution over future states and computes an expected value across it, robust analysis weakens this presumption, recognising that under deep uncertainty the distributions themselves are challenged [3].

The Robust Decision Making (RDM) framework developed by Lempert formalises this

distinction [5]. RDM evaluates strategies across thousands of plausible futures, identifies the conditions under which each strategy fails, and uses scenario discovery to characterise vulnerability. Haasnoot’s Dynamic Adaptive Policy Pathways approach extends this further by representing decisions as actions triggered by indicators in the evolving state of the world, producing adaptive policy maps rather than static recommendations. Fletcher argued that explicit modelling of climate-uncertainty learning over an infrastructure lifetime changes the ranking of flexible versus traditional designs [9]. Fletcher’s framing is influential but assumes that climate-uncertainty learning is exogenous to the planner; the present work treats demand uncertainty as resolved through realisation rather than through formal Bayesian learning, a deliberate simplification that loses the value of information explicitly but preserves transparency for stakeholder-facing tools.

Robustness is particularly developed in the water sector. Borgomeo et al. applied vulnerability-based stress-testing to UK water resource systems, identifying combinations of climate and demand stress under which existing portfolios fail [4]. Wang et al. developed a sequential decision-making framework with explicit uncertainty quantification for groundwater management [19]. Mahdi reviewed the broader Decision-Making under Deep Uncertainty (DMDU) literature in water planning, concluding that adoption in operational planning remains limited [12]. Hurford et al. show that diversified portfolios tend to perform robustly even when climatic uncertainty is overshadowed by institutional and financial variability — highlighting the importance of cross-sectoral uncertainty rather than climate alone [20]. Yet, RDM and DMDU frameworks can be computationally intensive and difficult to communicate to stakeholders, and integrated tools that combine flexibility with robustness remain rare. Brugnach et al. extended this critique, arguing that water-uncertainty research has been excessively focused on improving the precision of scientific data and predictive models at the expense of acknowledging the social and relational nature of the underlying decisions [21].

2.3 Multi-criteria Decision Frameworks

Flexibility and robustness usually evaluate design along a single financial axis. For long-lasting public infrastructure this is insufficient. Desalination plants have environmental, social and operational impacts that do not reduce to a single discounted cashflow. Multi-criteria Decision Analysis (MCDA) provides the methodology for structured evaluation across diverse criteria [22][23].

MCDA encompasses multiple methods: weighted-sum aggregation, the Analytic Hierarchy Process (AHP), outranking methods, and goal-programming approaches. However, these methods differ in their treatment of criterion disparity, the derivation of weights, and the handling of preference uncertainty. A substantial literature applies MCDA to water and, to a lesser extent, desalination decisions. Hajkowicz et al. reviewed 113 MCDA applications in water management, finding weighted-sum approaches and AHP to be the most common [24]. Makropoulos et al. developed an integrated MCDA-based decision-support system for sustainable option selection in urban water management, the closest stylistic precedent for the present work’s decision-support tool [25]. Makropoulos et al.’s system remains the closest stylistic precedent for an MCDA-anchored decision-support interface, but it pre-dates the integration of stochastic ROA outputs; the present DST extends the precedent by exposing Monte Carlo distributions and weight sensitivities, not only deterministic criterion scores. MCDA has been previously applied to desalination technology selection by Afgan et al., integrating economic, environmental, and social indicators [26].

Two limitations of standard MCDA practice are relevant here. First, MCDA is typically applied to a small set of pre-specified design alternatives, evaluated against fixed criterion

values. This is incompatible with the stochastic, scenario-ensemble framing of ROA and RDM, in which each candidate design produces a distribution of outcomes rather than a single value per criterion. Integration of MCDA with stochastic simulation requires either pre-aggregation (computing expected criterion values across the ensemble and feeding these to MCDA) or stochastic MCDA (preserving distributions through the multi-criteria aggregation step) [14]; the former is more common and is the approach adopted here. Second, criterion weights elicited from stakeholders are themselves uncertain, and the sensitivity of MCDA conclusions to weight perturbations is often underexplored. The present work addresses this through explicit weight-sensitivity analysis exposed through user-adjustable weight sliders in the Decision-Support Tool (Section 5).

A separate but related literature addresses the human-factors dimension of how multi-criteria information is presented to decision-makers. Comes et al. examined the cognitive load imposed by MCDA outputs on infrastructure planners and found that aggregated rankings frequently obscured the trade-offs that motivated stakeholders to commission the analysis [27]. Tsoukiàs [28] argued that decision-support tools should preserve the visibility of trade-offs through interactive visualisation rather than collapsing them into single composite indices. These findings motivate the design of the Decision-Support Tool described in Section 5, which exposes the underlying NPV distributions, per-axis MCDA scores, and weight sensitivities to the user rather than presenting only an aggregated ranking.

The use of MCDA in desalination planning has lagged its use in water-resource planning more broadly. Tariq et al. applied AHP to SWRO siting in Kuwait [29]. Cruz-Perez et al. used PROMETHEE to compare desalination configurations in the Canary Islands [30]. Neither study coupled the MCDA layer to a stochastic ROA framework: the design alternatives were treated as deterministic point estimates rather than as probabilistic distributions arising from underlying uncertainty. This decoupling is symptomatic of the broader integration problem the present work addresses.

2.4 Research Gaps

The literature shows clear benefits of applying flexibility, robust decision-making, and multi-criteria decision analysis to desalination or water infrastructure, but their integration remains incomplete in ways that limit their usefulness for SWRO capacity expansion planning. Several gaps remain unaddressed to fully realise strategic engineering’s potential.

- Gap 1: Existing studies typically apply ROA, robust assessment, and MCDA separately, leaving no unified method to evaluate economic, environmental and operational performance under deep uncertainty.
- Gap 2: The methodological refinements developed in other industries have not been systematically transferred to SWRO. The existing desalination-flexibility literature treats trigger rules and stochastic processes in a simplified form and does not address the energy-price-volatility issue Ghaffour (2013) identified as the dominant under-treated risk [31].
- Gap 3: Current modular-expansion trigger rules are often specified as fixed utilisation thresholds applied without build lags, abandonment behaviour, and maximum-build-rate constraints.
- Gap 4: Frameworks combining ROA, robustness, and MCDA are typically presented in the literature as analytical artefacts. The translation of these frameworks into tools usable by stakeholders is rare.

The positioning of the present work against the most directly comparable studies is summarised

in Table 1.

Table 1: Research Gaps in the Existing Literature

Paper	ROA	Modular Design	MCDA	Semi-arid Climate	Desalination	Decision Tool
Zhang (2012)	✓	✓	✓	✓		
Fletcher (2019)	✓					
Kuznetsova (2019)	✓	✓	✓			partial
Pachos (2022)	✓	partial	✓			
Hurford (2020)			✓			
Mahdi (2023)		✓	✓	✓		
Najafi (2024)	✓	✓		✓	✓	
This Work	✓	✓	✓	✓	✓	✓

2.5 Research Approach

The gap identified above is addressed by the methodology developed in Section 3, the case-study analysis in Section 4, and the Decision-Support Tool presented in Section 5. The methodological inheritance is from the cross-sector ROA frontier, primarily from Kuznetsova (2019) for the integrated decision-support structure and Bigestans (2023) for the build-lag and abandonment treatment of the trigger rule, and extended with a four-axis MCDA layer and a stakeholder-facing tool.

3 Methodology

This section develops the integrated framework that addresses the four gaps identified. The framework combines a stochastic simulation engine, a Real Options analysis, a robustness assessment protocol, and a multi-criteria decision layer.

3.1 Framework Overview

The framework is structured in four phases, as illustrated in Figure 1. Phase 1 is a stochastic simulation engine that propagates uncertain demand and electricity-price trajectories through a discounted cash-flow of the plant. Phase 2 applies this engine to value flexibility, comparing the expected NPV of a candidate flexible design against a rigid expansion across paired Monte Carlo trials. Phase 3 characterises the distributional robustness of each design through downside and upside percentile metrics and deterministic sensitivity analysis. Phase 4 combines the outputs across four criteria: economic, environmental, operational, and predictability.

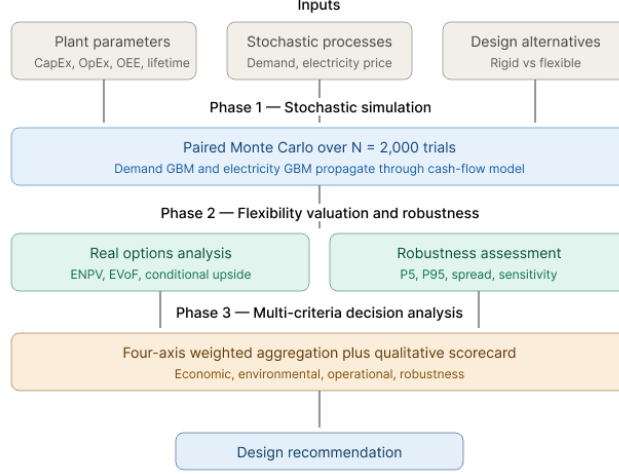


Figure 1: Four-Phase Framework Overview

The phased structure addresses the integration gap directly by ensuring ROA, robustness assessment, and MCDA share a single stochastic ensemble, rather than being computed independently and reconciled afterwards. The approach inherits from Kuznetsova’s integrated decision-support methodology, extending it across a four-axis MCDA layer and a stakeholder-facing tool. The framework followed is method-agnostic, with it being applied to SWRO desalination in this thesis to fix the parameter choices, the criterion set, and the case-study calibration, but the computational architecture is transferable.

3.2 Phase 1: Stochastic representation of uncertainty

The first phase of the framework models two stochastic state variables: annual water demand Q_t and the wholesale electricity price P_t^E . Both follow log-normal Geometric Brownian Motion (GBM) with the Itô correction, discretised to annual steps:

$$X_{t+1} = X_t \cdot \exp\left[\left(\mu_X - \frac{1}{2}\sigma_X^2\right) + \sigma_X \varepsilon_t\right], \quad (1)$$

with $\varepsilon_t \sim \mathcal{N}(0, 1)$ independent standard-normal Wiener increments and $\Delta t = 1$ year. This matches the discretisation used by Najafi (2024) in the comparable ROA literature, and the exponential form guarantees strict positivity. The drift parameters μ_X are log-returns; for parameters reported in source literature as arithmetic expected growth rates, the log-drift is computed via $\mu_X = \ln(1 + \mu_X^{\text{arith}})$.

The water-demand process applies Equation 1 directly. The electricity-price process applies the same form with an absorbing lower floor, reflecting regulatory and capacity-market mechanisms that bound day-ahead prices before they reach zero; the floor is binding in a negligible fraction of trials and serves principally as a numerical-stability safeguard. No equivalent floor is needed for demand, since the log-form discretisation guarantees $Q_t > 0$.

The framework supports user-selectable parameter calibrations for both processes, with drift and volatility independently configurable. This dual-regime exposes the parameter-uncertainty surface to the user rather than committing to a single calibration choice. The specific regimes, parameter values, and the active selection used in the SWRO case are presented in Section 4.

The demand and electricity-price increments are assumed independent: shocks to one do not propagate to the other. In reality the two variables share weak common drivers through macroeconomic conditions, but independence is adopted as a deliberate simplification in the absence of a robust correlation estimate at annual frequency.

For each Monte Carlo run, a single matrix of standard-normal increments $\{\varepsilon_t^{(j)}, \eta_t^{(j)}\}_{t=1, j=1}^{T, N}$ is drawn once and used to simulate both the rigid and the flexible design. Both designs are therefore evaluated against the same set of N futures, rather than against two independently-drawn sets. This is the Common Random Numbers technique: it ensures that the Expected Value of Flexibility reflects the structural difference between designs rather than sampling noise.

3.3 Phase 2: Real options analysis and flexibility valuation

The second phase values the flexibility embedded in the modular design by comparing it against the rigid design across the paired Monte Carlo ensemble generated in Phase 1. The flexible design embeds a sequential modular-expansion option: at each annual decision point, the planner observes realised demand and capacity utilisation and may, conditional on a trigger rule, commit to building an additional capacity module.

The option is valued by the simulation-based approach of deNeufville (2011): in each Monte Carlo trial, the rigid and flexible designs are simulated against the same stochastic future, and their Net Present Values are recorded. The trigger rule is heuristic, ensuring that flexibility is exercised only in response to genuine demand signals and always at a known cost.

The central financial output is the Expected Value of Flexibility (EVoF), defined as the expected difference in NPV between the two designs across the Monte Carlo ensemble. A positive EVoF indicates that the flexible design dominates the rigid design in expectation. EVoF is reported alongside the probability of flexible dominance and the conditional upside and downside, which together characterise both the size and the asymmetry of the option's payoff. The trigger rule is heuristic rather than optimal, so the reported EVoF is a lower bound on the value an optimal exercise policy would achieve.

The trigger rule structure, the cash-flow model, the discount rate, and the cost-architecture treatment of the two designs are presented in Section 4.

3.4 Phase 3: Robustness assessment

The third phase characterises the distribution of outcomes each design produces, rather than only its expected value. Two designs with identical Expected NPV may produce very different downside risk, and a planner is rarely indifferent between them. The framework therefore reports distributional metrics alongside the expected-value outputs of Phase 2, and supplements the stochastic ensemble with a deterministic sensitivity analysis to characterise robustness to non-stochastic parameters.

For each candidate design, the Monte Carlo ensemble produces a full distribution of Net Present Values across the N simulated futures. The framework summarises this distribution through three metrics: the median NPV, the 5th and 95th percentiles, and the standard deviation. The percentile spread $P_{95} - P_5$ is reported as the headline predictability metric: a tighter spread corresponds to a more predictable design. The framework also reports the fraction of trials in which each design outperforms the other, characterising whether the expected-value difference is driven by a consistent advantage or by a small number of extreme outcomes.

A separate one-at-a-time tornado analysis characterises robustness to deterministic parameters: each parameter is perturbed above and below its base value, with all other parameters held fixed and the stochastic engine disabled, and the resulting change in the Expected Value of Flexibility is recorded. A complementary flip-case analysis identifies parameter perturbations that reverse the sign of EVoF — the values at which the verdict between the two designs changes. The specific parameters tested, the perturbation magnitude, and the base values are

presented in Section 4.

3.5 Phase 4: Multi-criteria decision analysis

The fourth phase aggregates the outputs of the previous phases across multiple decision criteria. The financial metrics of Phase 2 and the distributional metrics of Phase 3 characterise design performance along a single monetary axis; for desalination infrastructure, performance along additional dimensions, such as environmental, operational, and risk-related, is also relevant to the investment decision.

For each candidate design, the Monte Carlo ensemble produces expected values of each decision criterion. These raw values differ in units and scale, so each is transformed into a comparable score through floor-based linear scaling: the worst-performing design on a criterion receives a baseline score, the best receives the maximum, and intermediate values are interpolated. The baseline floor prevents small numerical differences from being visually exaggerated on a fixed scale.

The transformed scores are then combined into a composite through weighted-sum aggregation, with user-adjustable criterion weights summing to one. Weighted-sum is selected over alternatives because it is the simplest defensible aggregation, is the most common method in published water-MCDA applications [24], and exposes criterion contributions transparently to non-specialist users.

The specific criteria, weights, and scaling parameters are presented in Section 4.

4 Case Study: Expansion of the Torrevieja SWRO Desalination Plant

This section applies the proposed methodology to a representative SWRO case: the Torrevieja desalination plant on the Spanish Mediterranean coast featured in Figure 2a, currently operating at 80 hm³/yr capacity with a planned expansion of +40 hm³/yr. While the case fixes parameter values, calibration sources, and decision-rule specifics, the framework itself remains transferable to other SWRO contexts.

4.1 Plant context and the expansion question

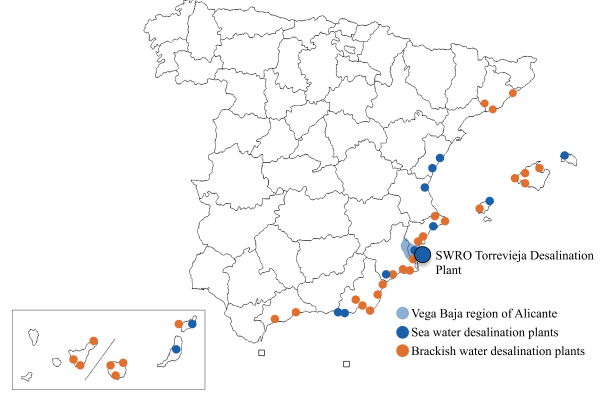
The Torrevieja plant was commissioned in 2014 as one of the largest desalination plants in Europe. It supplies drinkable and agricultural water across the southern Vega Baja region of Alicante as illustrated in Figure 2b. The plant is operated by Acuamed under a Spanish state concession and serves a service area whose demand has grown faster than original forecasts anticipated, prompting an active expansion programme to raise total capacity by 50% [32].

The expansion can be implemented in two structurally different ways. The first is a single staged rigid build: the full +40 hm³/yr capacity committed upfront with construction phased across three years, in agreement with the proposed timeline. The second is a modular incremental expansion: capacity added in desalination modules of 5,000 m³/day, with each module triggered only when realised demand justifies it. The two designs are equivalent in total capacity but differ fundamentally in how capacity is committed over time, and therefore in how each responds to demand realisations that diverge from the forecast.

Torrevieja is a suitable case for the framework for three reasons. The expansion decision is live and economically material at the time of writing, rather than retrospective. Operational



(a) SWRO desalination plant in Torrevieja.
(Source: Acuamed [33])



(b) Spain's desalination plants locations

Figure 2: SWRO Torrevieja Desalination Plant context

data is available in the public record at sufficient resolution to anchor the most influential parameters of the stochastic processes; remaining parameters are calibrated against industry benchmarks or assumed explicitly where authoritative sources are unavailable, with each assumption documented in the relevant section below. And the Mediterranean climate context exposes the plant to the demand-and-energy uncertainty regime the framework is designed to address.

4.2 Plant alternatives and cost model

Capital expenditure for the rigid design is distributed across three construction years in a 25%/35%/40% pattern, reflecting site preparation and intake works in the first year, mechanical and electrical installation in the second, and commissioning in the third. The flexible design replaces this single staged outlay with per-module expenditure: 22 identical modules of 5,000 m³/day, with each module's unit cost set at €3.0 M reflecting modular SWRO train procurement at industry pricing. Operating expenditure separates a fixed component from a non-energy variable component (chemicals, membrane replacement, labour), with electricity treated separately as the dominant variable cost and the only line subject to stochastic uncertainty. Table 2 summarises the cost stack used in both designs.

The flexible per-module CapEx implies a maximum total expansion outlay of $22 \times €3.0 \text{ M} = €66 \text{ M}$, lower than the €89 M committed by the rigid design. This reflects two compounding effects in modular procurement: avoidance of the integration and overdesign overhead embedded in a single monolithic build, and competitive market pricing for standardised 5,000 m³/day SWRO trains [34]. The differential is conservative: in practice, smaller-scale modular SWRO units typically incur a per-cubic-metre premium of 10–20% relative to monolithic capacity due to lost economies of scale in pretreatment and energy recovery; the absence of such a premium in the present cost model is documented as a structural choice that biases the comparison toward the modular design and is interrogated in the sensitivity analysis of Section 6.3.

The non-energy variable cost differs between the two designs because each flexible module is commissioned with newer-generation membranes and energy-recovery devices than would have been installed in a single upfront rigid build. The full per-component cost decomposition is presented in Appendix 9.3.

Table 2: Cost components for the rigid and flexible designs.

Cost component	Value	Notes
Total rigid CapEx	€89 M	Staged 25%/35%/40% across years 0–2
Flexible module unit CapEx	€3.0 M	Per-module procurement at modular-train pricing
Flexible maximum total CapEx	€66 M	$22 \times €3.0$ M, demand-tracking
Fixed OpEx	€10 M/yr	Scales with installed capacity
Rigid non-energy variable cost	€0.20/m ³	Chemicals, membrane replacement, labour
Flexible non-energy variable cost	€0.14/m ³	Newer-vintage membrane efficiency
Energy intensity	4.5 kWh/m ³	SWRO industry standard [34]

4.3 Stochastic calibration: demand and electricity price

The framework supports two source-anchored calibrations for each stochastic process, with drift and volatility independently selectable in the Decision-Support Tool (Section 5). This section presents both regimes, justifies the active mix-and-match selection used for headline reporting in Section 6, and provides the empirical rationale for each parameter choice.

4.3.1 Water demand

Two calibration regimes are available, summarised in Table 3. The historical calibration estimates drift and volatility directly from the 2003–2025 record of Spanish coastal water demand [35][36]. The literature calibration takes the drift from forward-looking European Commission projections of desalinated water demand under climate stress [37], and sets volatility at a central estimate consistent with the volatility ranges adopted in the similar ROA literature for long-horizon demand uncertainty (Cardin (2015) use 5–12%; Bigestans (2023) uses 8–15%).

Table 3: Water-demand calibration regimes.

Regime	μ_Q^{arithh} (%/yr)	σ_Q (%/yr)	Source
Historical	8.33	29.1	MCT and Seguridad Hídrica report
Literature	7.00	15.0	European Commission projections

The historical volatility of 29.1%/yr is inflated by three structural breaks spanning the calibration window. The AGUA Programme (*Actuaciones para la Gestión y la Utilización del Agua*, 2004–2010) accelerated coastal desalination construction, producing supply-side step changes that propagated into observed demand. The 2008 financial crisis suppressed industrial and agricultural demand for several years. The 2012 and 2017 drought episodes drove transient demand spikes. Each is a regime-specific event rather than a sample of the underlying process, and together they inflate the historical variance relative to what the next fifteen years are likely to produce.

4.3.2 Electricity price

Two calibration regimes are also available for electricity, summarised in Table 4. The historical calibration estimates drift and volatility from OMIE 2010–2025 Spanish wholesale day-ahead prices [38]. The literature calibration retains the historical drift estimate but reduces volatility to a central-estimate value consistent with the volatility ranges adopted in the comparable ROA literature [39], [40].

The elevated historical volatility of 21.9%/yr is dominated by a single structural event: the

Table 4: Electricity-price calibration regimes.

Regime	μ_E (%/yr)	σ_E (%/yr)	Source
Historical	1.23	21.9	OMIE and Datos.macro 2010–2025
Literature	1.23	12.0	Comparable ROA literature

2021–2022 European energy crisis, in which gas supply constraints following the Russian invasion of Ukraine drove Spanish wholesale prices to several times their pre-2020 norms before regulatory intervention returned the market toward equilibrium. This episode is a regime-specific break, not a sample of the underlying process volatility SWRO planners are likely to face across the 2025–2040 horizon.

The absorbing lower floor on the electricity-price process is set at $P_{\min}^E = \text{€}0.01/\text{kWh}$. The value reflects the empirical observation that wholesale prices in the Spanish day-ahead market are bounded below by capacity-market and regulatory mechanisms before reaching zero; it serves principally as a numerical-stability safeguard.

4.3.3 Active calibration selection

Across both stochastic processes, the same selection logic applies: historical drifts capture empirically defensible base rates, while historical volatilities are partly artefacts of regime-specific events not representative of the next fifteen years. Table 5 summarises the resulting active selections and their rationale.

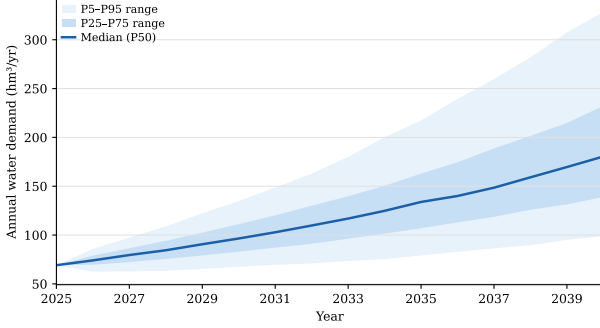
Table 5: Active stochastic parameter selections and calibration regimes

Stochastic process	Parameter	Symbol	Value	Underlying calibration logic
Water demand	Drift	μ_Q	8.33%/yr	Historical base rate provides an empirically defensible trend projection for the 15-year horizon.
	Volatility	σ_Q	15.0%/yr	Literature estimate filters out regime noise and anomalies (crisis, structural drops) from historical files; central value within the 8–15%/yr range adopted by comparator ROA studies.
Electricity price	Drift	μ_E	1.23%/yr	Historical base rate captures long-run real price trends under generalised market normalisation.
	Volatility	σ_E	12.0%/yr	Literature estimate excludes temporary geopolitical and extreme supply shocks (e.g., 2021–2022 energy crisis).

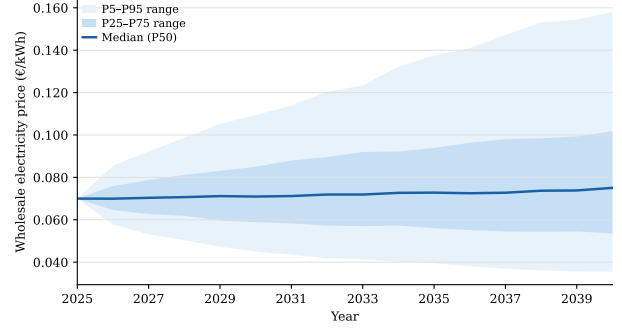
The literature volatilities are also directly comparable to those adopted in similar comparator ROA studies, supporting cross-study interpretability of the Expected Value of Flexibility reported in Section 6 [39][40].

The combined effect of the active calibration on the simulated future is illustrated in Figure 3, which shows the central tendency and uncertainty bands across the 2,000-trial Monte Carlo ensemble for both stochastic processes. The water-demand fan widens steadily through the 15-year horizon, reflecting the compounding of the moderate Literature volatility on the

Historical drift. The electricity-price fan widens more sharply in absolute terms but from a much lower starting value, and its central line remains close to the initial price as the modest 1.23%/yr drift is largely offset by the symmetric volatility around it. Together the two fans visualise the uncertainty surface against which the rigid and flexible designs will be evaluated in the cash-flow model (Section 4.4) and the simulation (Section 6).



(a) Water demand



(b) Wholesale electricity price

Figure 3: Monte Carlo fan charts of the two stochastic processes under the active calibration over the 15-year horizon.

4.4 Decision-rule parameters

The flexible design exercises its sequential expansion option through the trigger rule introduced in Section 3.3: at the end of each year, the planner computes a two-year rolling utilisation and either commits a new module, abandons further expansion, or takes no action as illustrated in Figure 4. This section specifies the parameter values that operationalise the rule for the Torrevieja-scale case, summarised in Table 6.

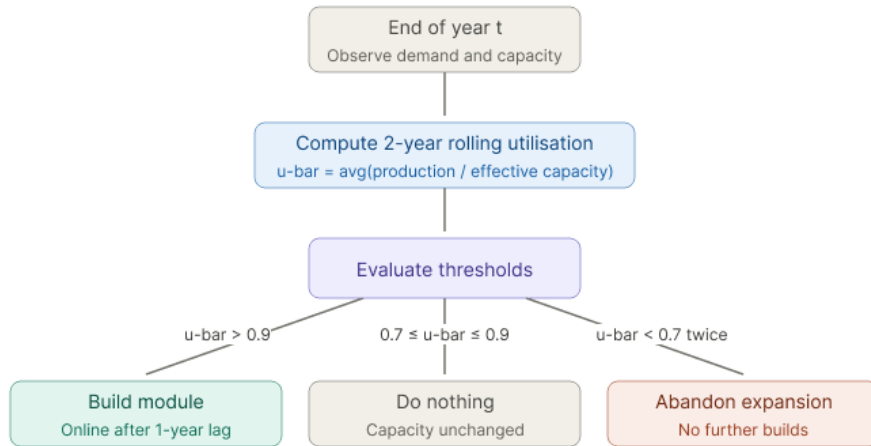


Figure 4: Expansion trigger rule

The 0.85 build threshold reflects that expansion is warranted as the plant approaches design utilisation under continuing growth; the slightly lower value relative to a 0.90 threshold provides additional lead time for the one-year build lag to deliver capacity before utilisation reaches binding levels. The 0.70 abandonment threshold, evaluated over a two-year window, distinguishes a structural demand shortfall from a single-year low realisation.

The 1-year build lag matches typical SWRO module construction and commissioning timelines. The 5,000 m³/day module size is the standard commercial SWRO train scale and corresponds to the unit cost of €3.0 M in Section 4.2. The 22-module ceiling sets the total expansion

Table 6: Decision-rule parameters for the flexible expansion programme.

Parameter	Value	Notes
Build threshold (u_{build})	0.85	Module ordered when rolling utilisation exceeds 85%
Abandonment threshold (u_{abandon})	0.70	No further builds if utilisation falls below 70% for two consecutive years
Build lag (L_b)	1 year	Construction and commissioning time per module
Module size	5,000 m ³ /day	Standard commercial SWRO train size
Maximum module count (M_{max})	22	Sets the upper expansion envelope to +40 hm ³ /yr
Maximum builds per year	10	Caps construction throughput
Operating efficiency (rigid)	85%	Reflects rigid plant utilisation losses
Operating efficiency (flexible)	90%	Reflects newer-module efficiency advantages

envelope at $22 \times 5,000 \times 365 \approx 40$ hm³/yr, matching the rigid plant’s nameplate addition.

The two designs operate at different effective availability levels. The rigid plant carries an 85% OEE typical of large monolithic SWRO facilities, where stoppages cascade across the full capacity envelope. The flexible plant carries a 90% OEE reflecting modular maintenance scheduling and the newer membrane and energy-recovery technology installed at each commissioning.

The OEE differential and the non-energy variable-cost differential (€0.20 vs €0.14/m³) are *technology-vintage* effects: they capture that modular capacity installed across years 3–8 will, in expectation, use newer membranes, more efficient energy-recovery devices, and benefit from staggered-maintenance scheduling unavailable to a monolithic 2025-vintage commissioning. They are conceptually distinct from the *option-timing* value of being able to defer, expand, or abandon capacity in response to demand realisations. The Expected Value of Flexibility reported in Section 6 should therefore be interpreted as the *combined* advantage of the modular design — option-timing value plus technology-vintage advantage — rather than as a pure option premium. The combined framing is defensible because it is the comparison a real planner faces: rigid commitment to 2025-vintage technology against staged commitment that captures vintage improvements as a structural feature of modularity. A decomposition simulation isolating the two components is identified as a priority extension in Section 7.3.

4.5 Financial model and evaluation

The framework computes the Net Present Value of each design over a 15-year horizon, applying a discount rate of $r = 8\%$ that reflects the weighted-average cost of capital for desalination water infrastructure [41]. This section specifies the three cash-flow components of the model in turn (revenue, variable cost, and capital expenditure) and then defines the financial valuation metrics derived from them.

4.5.1 Revenue

Annual revenue is calculated by multiplying realised production by its regulated water tariff:

$$R_t^{d,(j)} = P_t^W \cdot \text{Production}_t^{d,(j)}, \quad (2)$$

where the water tariff is $P_t^W = €1.30/\text{m}^3$ in year zero and drifts forward at the electricity-price drift rate to reflect pass-through of energy costs into regulated tariffs. Production is bounded

above by effective capacity: $\text{Production}_t^{d,(j)} = \min(Q_t^{(j)}, \text{Capacity}_t^d \times \text{OEE}^d)$. The rigid design enters year 3 at full $80 + 40 = 120 \text{ hm}^3/\text{yr}$ nameplate capacity; the flexible design's capacity grows step-wise as modules are committed by the trigger rule.

4.5.2 Variable cost

Variable cost, $C_{\text{var},t}^{d,(j)}$, combines a non-energy component, c_v^d , and an energy component, $e \cdot P_t^{E,(j)}$:

$$C_{\text{var},t}^{d,(j)} = c_v^d \cdot \text{Production}_t^{d,(j)} + e \cdot P_t^{E,(j)} \cdot \text{Production}_t^{d,(j)}, \quad (3)$$

with c_v^d taking the design-specific values from Table 2 and energy intensity $e = 4.5 \text{ kWh/m}^3$. The energy component is the only cash-flow line directly exposed to stochastic price variation through $P_t^{E,(j)}$.

4.5.3 Capital expenditure

The two designs differ structurally in CapEx. The rigid design commits the full €89 M staged across years 0–2 in the 25%/35%/40% pattern, independent of the realised demand trajectory. The flexible design commits €3.0 M per module at each commitment date determined by the trigger rule, whilst CapEx in years where no module is committed is zero. The combined effect is that the rigid design has front-loaded, deterministic CapEx, while the flexible design has demand-tracking, stochastic CapEx that matches capital commitment more closely to realised demand.

4.5.4 Net Present Value (NPV)

For each Monte Carlo trial j and candidate design $d \in \{\text{rigid}, \text{flex}\}$, the Net Present Value aggregates the three cash-flow components and fixed operating expenditure into a discounted total:

$$\text{NPV}_d^{(j)} = \sum_{t=0}^T \frac{R_t^{d,(j)} - C_{\text{var},t}^{d,(j)} - \text{OpEx}_{\text{fix}}^d - \text{CapEx}_t^{d,(j)}}{(1+r)^t}, \quad (4)$$

where $R_t^{d,(j)}$ represents the revenue, $C_{\text{var},t}^{d,(j)}$ is the variable production cost, and $\text{OpEx}_{\text{fix}}^d$ scales with installed capacity at the values introduced in Section 4.2.

4.5.5 Flexibility Valuation

The headline financial output of the framework is the Expected Value of Flexibility (EVoF), the difference in expected NPV between the two designs across the Monte Carlo ensemble:

$$\text{EVoF} = \text{ENPV}_{\text{flex}} - \text{ENPV}_{\text{rigid}}. \quad (5)$$

A positive EVoF indicates that the flexible design dominates the rigid design in expectation. The interpretation is unambiguous because paired sampling ensures both designs are evaluated against the same set of N futures: the difference reflects the structural value of the embedded expansion option, not the noise of independent samples. EVoF is reported alongside the probability of flexible dominance and the conditional upside and downside, which together characterise both the size and the asymmetry of the option's payoff. The trigger rule is heuristic rather than optimal, so the reported EVoF is a lower bound on the value an optimal exercise policy would achieve.

4.6 Multi-criteria decision analysis: criterion specification

The quantitative MCDA layer evaluates each design across four criteria. For each criterion, the Monte Carlo ensemble produces an expected value per design, which is transformed to a comparable 0–100 score through floor-based linear scaling:

$$S_{d,k} = 15 + 85 \cdot \frac{I_{d,k} - I_k^{\min}}{I_k^{\max} - I_k^{\min}}, \quad (6)$$

for benefit-type indicators, with the symmetric form for cost-type indicators. The floor of 15 reflects the observation that both designs are viable plants; it prevents small numerical differences from being visually exaggerated on a fixed scale. The scaled scores are aggregated into a composite:

$$S_d = \sum_{k=1}^4 w_k S_{d,k}, \quad \sum_{k=1}^4 w_k = 1, \quad (7)$$

with user-adjustable criterion weights, w_k . Table 7 specifies the operationalisation, scaling direction, and base-case weight for each criterion.

Table 7: MCDA criterion specification and base-case weights.

Criterion	Operationalisation	Direction	Base-case weight
Economic	Mean NPV across 2,000 trials	Maximise	25%
Environmental	Mean fraction of demand met across 2,000 trials	Maximise	25%
Operational	Composite of mean utilisation (50%) and CapEx schedule certainty (50%)	Maximise	25%
Predictability (NPV dispersion)	Inverse of P95–P5 NPV spread	Maximise	25%

Supply reliability serves as a proxy for direct environmental impact on the grounds that unmet demand drives load onto more carbon-intensive marginal sources. The proxy is appropriate for comparing two designs at identical per-cubic-metre loadings and operating profiles.

The operational axis is a deliberately competitive combination. The flexible design wins on utilisation: modular deployment right-sizes capacity to demand, avoiding systematic under-utilisation in low-demand trials. The rigid design wins on CapEx certainty: its construction programme is fully determined in advance and carries zero trial-level variance. Weighting both equally ensures the composite genuinely discriminates between the two designs rather than rewarding either twice.

The predictability axis is a deliberately narrow operationalisation of robustness as outcome *predictability*, measured through NPV dispersion. An alternative operationalisation as downside protection (the P5 floor) would reverse the verdict, since the flexible design’s P5 outperforms the rigid design’s (Section 6.2). The axis label has been chosen to make this scope explicit: a stakeholder concerned with worst-case outcomes should down-weight Predictability and consult the P5 floor directly.

Equal base-case weights ($w_k = 0.25$) reflect a neutral starting point in the absence of elicited stakeholder preferences. The weights are user-adjustable in the Decision-Support Tool, and

weight-sensitivity analysis is reported in Section 6.

5 Decision-Support Tool

The framework developed in the preceding sections is delivered as an interactive web-based interface. This section presents its design objectives, architecture, and interface, the human-factors rationale behind its key design decisions, and the usability evaluation conducted with two distinct user groups.

5.1 Motivation and design objectives

In its native form, the framework is a simulation model accessible only to a user fluent in stochastic modelling and spreadsheet computation. The Decision-Support Tool is designed to make that model operational for two related audiences. Its primary users are specialists, such as analysts, engineers, and infrastructure planners who produce or actively interrogate stochastic investment analyses. Its secondary users are the non-technical stakeholders to whom those specialists communicate: planners, committee members, and public decision-makers who consume the resulting recommendations and must understand, interrogate, and act on them without specialist training. The gap between the production of a flexibility analysis and its communication to the latter group is identified in the water-planning literature as a primary barrier to the operational uptake of decision-making-under-uncertainty methods [12].

The tool is designed to serve both audiences through a single interface. It adds no analytical capability beyond the underlying model. Its purpose is to make the model's outputs both rigorous for specialists and legible for non-specialists. Four objectives follow: to expose uncertainty rather than conceal it, to make the multi-criteria trade-off visible, to let the user adjust the assumptions that matter to them and see the consequences in real time, and to deliver a plain-language recommendation that non-technical stakeholders can act on and communicate. These objectives place the tool in the tradition of interactive decision-support systems that preserve the visibility of trade-offs rather than collapsing them into an aggregated ranking [25][28].

5.2 Architecture and technology

The tool is built using Python and deployed to Render.com. All plant assumptions are read once at startup from the Excel workbook; the Monte Carlo engine then runs 2,000 paired trials on demand whenever the user adjusts parameters. No installation is required: the tool runs entirely in the browser, accessible via a single URL (linked at the abstract).

The interface is organised across five tabs, each corresponding to a distinct stage of the decision process, separated into two sections (overview, and simulation and analysis):

1. **About:** research context, methodology overview, and navigation guide.
2. **Worked Example:** a single-scenario deterministic walkthrough for both designs.
3. **Inputs:** user-adjustable parameters. Pressing *Apply* re-runs the full 2,000-trial simulation and updates all downstream tabs.
4. **Uncertainty:** the stochastic Monte Carlo output.
5. **Decision:** a structured recommendation using plain-language.

5.3 Tool description

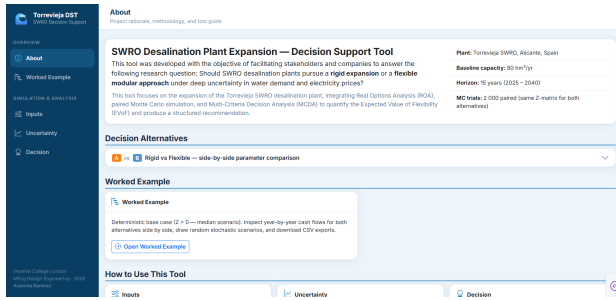
This section walks through the five tabs in the order a user encounters them, with screenshots of the three tabs that carry the most analytical content (Figures 6, 7a, and 7b).

5.3.1 Tab 1: About

The opening tab introduces the research context, the comparison being made (rigid staged expansion vs modular incremental expansion), and a brief methodology overview as illustrated in Figure 5a. It serves as an onboarding device: a user unfamiliar with real options or Monte Carlo analysis can read this tab before interacting with any parameters, reducing the cognitive burden of the first encounter with the tool.

5.3.2 Tab 2: Worked Example

The second tab presents a single deterministic scenario as a concrete walkthrough. It displays the year-by-year cash flows and capacity evolution for both designs under one randomly-drawn future, alongside headline KPIs (rigid NPV, flexible NPV, EVoF). The user can draw a new random scenario to see how the two designs respond differently to different demand realisations and read the expansion recommendation as illustrated in Figure 5b.



(a) Opening tab



(b) Single deterministic scenario

Figure 5: Overview interface of DST

5.3.3 Tab 3: Inputs

The inputs tab exposes the user-adjustable parameters: demand and energy drift and volatility (with regime presets for Literature and Historical), discount rate, water price, rigid and flexible OEE, variable costs, and build and abandon trigger thresholds. Pressing *Apply* re-runs the full simulation and updates Tabs 4 and 5. The regime selectors allow the mix-and-match calibration described previously: the user can independently select drift from one regime and volatility from another, observing the effect on the headline result in real time as illustrated in Figure 6.

5.3.4 Tab 4: Uncertainty

The uncertainty tab presents the full Monte Carlo output. Water-demand and electricity-price fan charts show the simulated P10–P90 bands with historical observations overlaid for context. An NPV histogram displays the overlapping distributions for both designs. A percentile table reports multiple possible NPVs for each design alongside the EVoF. A capacity timeline shows the P10–P90 band of cumulative modules built over the project horizon. An on-demand tornado chart computes the sensitivity of EVoF to each input parameter, with a stakeholder-readable summary identifying which parameters most affect the recommendation and whether any perturbation reverses the verdict as illustrated in Figure 7a.

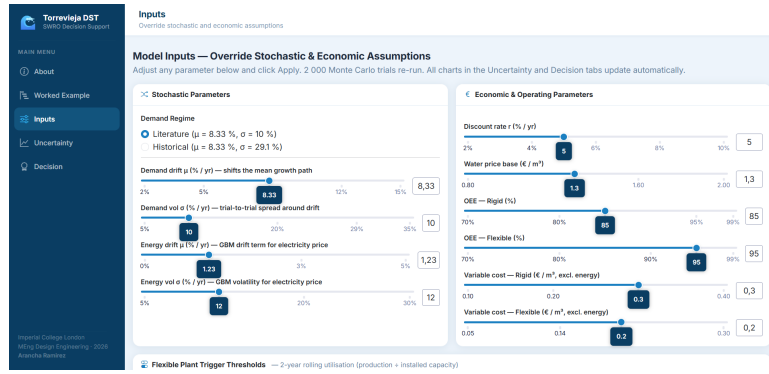
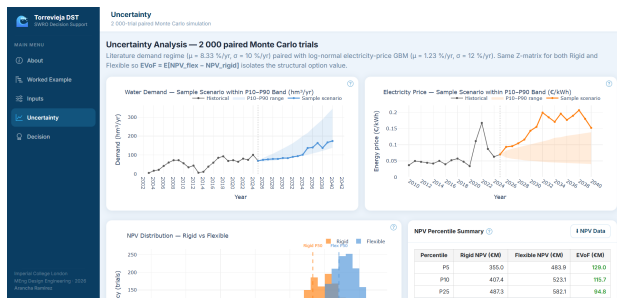


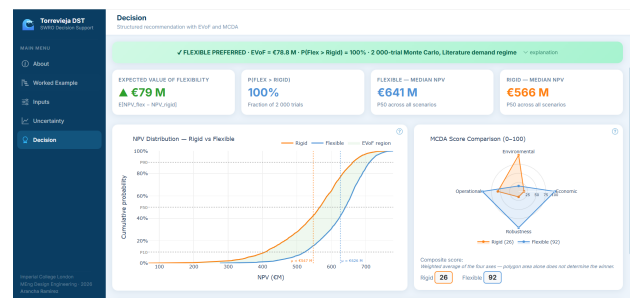
Figure 6: Inputs tab showing parameter sliders, and the Apply button that triggers a full 2,000-trial Monte Carlo re-run.

5.3.5 Tab 5: Decision

The decision tab synthesises the simulation output into a structured recommendation. A colour-coded banner delivers a verdict that expands into a non-technical explanation of the result. Headline KPI cards display the EVoF, the probability of flexible dominance, and the median NPV of each design. A paired CDF chart shows the empirical cumulative distributions, with the horizontal gap at any probability level representing the EVoF at that percentile. The MCDA radar chart scores both designs on the four axes (Economic, Environmental, Operational, Predictability) on a 0–100 scale, with the composite weighted score displayed beneath. Four weight sliders allow the user to adjust criterion importance and observe the composite score update immediately as illustrated in Figure 7b.



(a) Stochastic risk visualisation



(b) Strategy recommendation

Figure 7: Interface output of DST

5.4 Human-factors design choices

The tool's analytical depth is delivered through the underlying data layer described in the previous sections. The interface design choices in this section sit alongside that depth, lowering the entry threshold for non-specialist stakeholders while preserving full analytical access for specialists.

The decision tab opens with a colour-coded banner (green when flexibility is preferred, amber when it is not) containing a one-sentence plain-language verdict. Clicking the banner expands a short non-technical explanation of the underlying reasoning. The two-layer structure addresses the most common failure mode of decision-support tools: producing technically correct outputs that no one acts on. The banner delivers the conclusion at a glance for users who do not wish to interpret the full analytical output; the expandable detail provides the reasoning for those who want to interrogate it.

Throughout the interface, information icons (?) are placed alongside technical terms, chart

titles, and parameter labels. Hovering over an icon reveals a short tooltip explaining the concept in non-specialist language. This reduces the need for external documentation or prior training: a user encountering “EVoF” or “P90–P10 spread” for the first time can access a plain-language definition without leaving the tab. The tooltips are deliberately brief to avoid interrupting the workflow of experienced stakeholders for whom the terms are already familiar.

The five-tab structure is designed so that analytical complexity increases as the user moves through the tool. Tab 1 provides context. Tab 2 shows a single concrete scenario. Tab 3 exposes parameter controls. Tab 4 presents the full distributional output. Tab 5 delivers the synthesised recommendation. A user who reaches the recommendation has already been exposed to the uncertainty and trade-offs underlying it, rather than encountering a verdict without context. This progressive structure reduces the cognitive load of engaging with 2,000 simulated futures by staggering the user’s understanding one layer at a time. This is particularly useful for non-specialist users, who can engage with each layer at their own pace.

The second tab presents a single deterministic scenario before the user encounters the full Monte Carlo output. This addresses the unfamiliarity barrier: a user who has never seen a fan chart or an NPV distribution can first watch one future unfold year by year, seeing how each design responds to that specific trajectory, before being asked to reason about 2,000 futures simultaneously, whilst more experienced users can choose to skip this tab. The “draw new scenario” button lets the user see multiple individual trajectories and build intuition for the variability before moving to the distributional view.

5.5 Usability evaluation

The tool’s design objectives and human-factors choices are evaluated through a structured usability study with two distinct user groups, followed by a complementary semi-structured interview with a practising water-utility engineer. This section presents the evaluation methodology in full and reports findings from data collected to date.

5.5.1 Evaluation methodology

The tool is designed primarily for specialist users while aiming to lower the learning curve sufficiently for non-specialist stakeholders to engage with the same outputs. The evaluation tests both ends of this design intent. The protocol is positioned as a *formative* usability evaluation conducted under a low-risk human-subjects research approach: participants gave informed consent, no personally identifying information was retained, and no clinical, financial, or otherwise sensitive data was collected. A *summative* evaluation appropriate for confirmatory claims would additionally apply validated psychometric instruments such as the System Usability Scale (SUS) for perceived ease of use, NASA-TLX for cognitive load, and the User Experience Questionnaire (UEQ) for affective response. These were considered out of scope for an early-stage artefact whose primary purpose was to surface design signals for the next iteration; the lighter five-stage protocol described below was selected on those grounds.

Ten participants were recruited across two groups of equal size.

Group A (specialists, $n=5$): participants with background in one or more of data analysis, systems design, economics or financial modelling, or desalination construction and facility management. This group represents the primary user persona — the analyst or planner who produces or actively interrogates the analysis — and evaluates the technical credibility of the tool’s outputs and the analytical depth of its interface.

Group B (non-specialists, $n=5$): participants with no prior experience with decision-support tools, data analysis frameworks, or infrastructure financial modelling. This group represents

the secondary, non-technical stakeholder persona the tool is designed to remain accessible to — the planner, committee member, or public decision-maker who receives an investment recommendation and must understand, interrogate, and act on it without specialist training.

The two-group design enables an evaluation that operates at both ends of the design intent: Group A validates whether the tool functions as a primary working artefact for technically competent users, while Group B tests whether the layered design succeeds in extending accessibility to a non-technical audience.

Each participant completed a five-stage protocol conducted individually:

1. Introduction: The participant received a brief overview of the research context and the comparison being evaluated. No instruction was given on how to use the tool.
2. Free exploration: The participant was given unguided access to the tool for approximately 5–10 minutes, navigating at their own pace. The researcher observed but did not intervene.
3. Task-based evaluation: The participant was asked to complete a standardised set of tasks:
 - Locate the Expected Value of Flexibility and state which design is preferred.
 - Locate and explain the recommended expansion deployment plan.
 - Explain in their own words why the tool recommends the proposed design.

Task completion, hesitation points, and errors were recorded by the researcher.

4. In-person interview: A semi-structured interview explored the participant’s experience: what they found intuitive, what they found confusing, whether they trusted the recommendation, and what they would change.
5. Online form: After the session, the participant completed a short questionnaire capturing quantitative ratings (cognitive load, frustration, perceived task success, ease of use) and open-ended feedback.

5.5.2 Findings

Quantitative ratings are summarised in Table 8 and Figure 8a.

Table 8: Mean usability scores by participant group.

Metric (1–10)	Group A	Group B	Overall	Range
Cognitive load	7.0	5.8	6.4	3–8
Frustration	2.4	5.2	3.8	1–9
Perceived task success	8.0	4.2	6.1	1–10
Ease of use	7.6	7.0	7.3	4–10

The main insight from Table 8 is the asymmetry between groups. The specialist group reports high perceived task success (8.0/10), low frustration (2.4/10), and ease of use comparable to that of the non-specialist group (7.6 vs 7.0). This is direct evidence that the tool’s primary design objective, to function as an analytical artefact for specialist users, has been met. The non-specialist group reports substantially lower perceived task success (4.2/10) and higher frustration (5.2/10), indicating that the accessibility extension into the non-specialist audience is partial rather than complete. The tool succeeds at being operable for non-specialists but does not fully succeed at being legible to them.

The asymmetry above is most starkly visualised in Figure 8b: four of five Group B participants sit below the $y = x$ reference line, indicating that the tool was easier to navigate than to

interpret. The clearest expression of this pattern is the participant who rated ease-of-use at 10/10, navigation as Strongly Agree on every Likert axis, but task success at 1/10, commenting only “I don’t understand the numbers.” Another non-specialist reported that the flow was intuitive but they “didn’t understand the data and technical terms”. Navigability and comprehension are distinct achievements, and the evaluation suggests the tool secures the former more fully than the latter for the non-specialist audience.

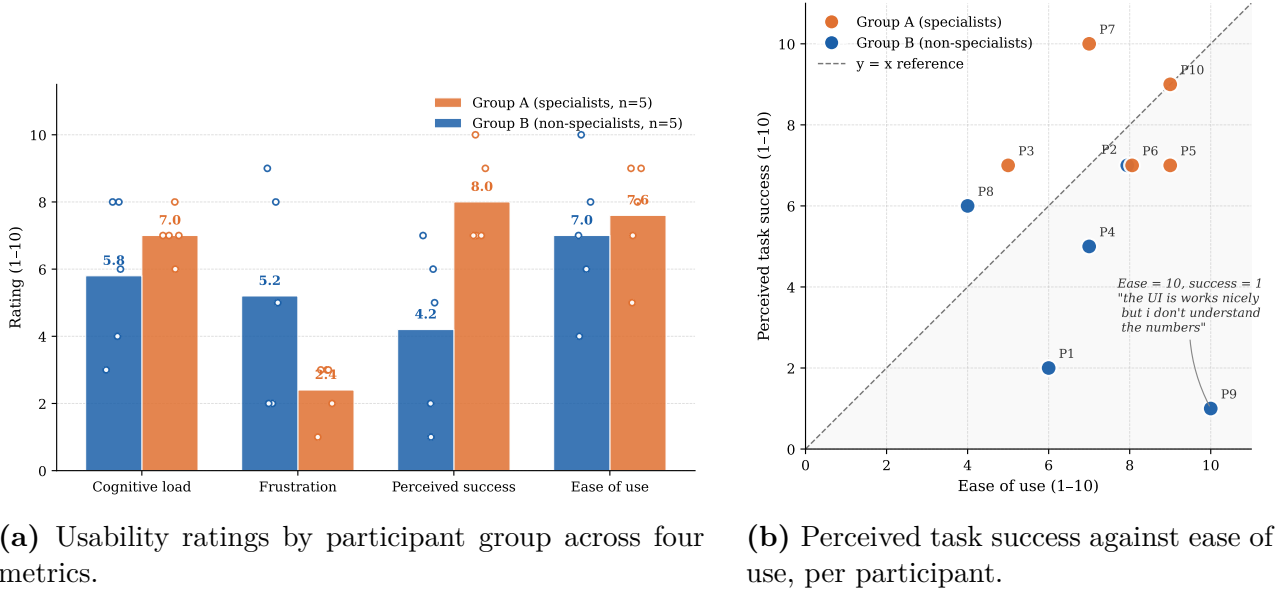


Figure 8: User testing responses

When asked which single visual component best communicated the risk difference, six of ten participants selected the Headline KPI Cards. The remaining selections distributed across the MCDA radar (one Group B participant), the NPV distribution (one Group A), and a freeform reference to the MCDA in combination with other elements (one Group A participant, who explicitly noted the radar was not updating during her session). The most direct, least graphical summary was the most effective at conveying the comparison, particularly for the non-specialist group, which selected the KPI cards in three of five cases. The more sophisticated visualisations (CDF, radar, capacity timeline) were preferred by specialists or chosen as part of a composite reading.

The hoverable information icons received explicit positive mention from multiple Group B participants. One non-specialist wrote that they “found the icons with question marks that explain them extremely useful” but wanted “more of them,” particularly for acronyms. The single most frequent feature request across the cohort was a glossary or dedicated help section defining technical terms. Several participants reached comprehension only after asking the researcher questions during the session, indicating that the existing tooltip provision did not cover the vocabulary that non-specialists found opaque.

Group A participants consistently asked for features oriented toward real-world workflow rather than comprehension. Three independently requested the ability to export, save, or share their inputs, weights, and results (as a CSV or as a shareable link) so that an analysis could be shared with colleagues or used in a presentation. One requested a distinction between a simplified default mode and an advanced mode, with saved input presets and personalisation by user profile. These requests indicate that specialists were evaluating the tool as a prospective workplace artefact, and that its current single-mode, session-bound design does not yet support collaborative or repeated professional use.

Several discrete issues were identified: one participant mistook a “+” icon for an interactive

button; one identified a contradictory sentence in the About tab describing the module deployment lag; one requested a reset control for the inputs tab; and two commented on the placement of the recommendation, one finding its position at the foot of the tab confusing and another suggesting that the primary decision be surfaced at the top of the interface.

5.5.3 Practitioner interview

The formative evaluation was supplemented by a semi-structured interview with a practising engineer at a national water utility in Southeast Asia. This interview is reported as complementary evidence rather than as part of the structured ten-participant study: the protocol was less formal, no quantitative ratings were collected, and the participant was not asked to perform the standardised task set. Its value lies in addressing the ecological-validity limitation of the main study — specifically, the absence of practising water-infrastructure planners from the participant pool.

The practitioner’s feedback aligned with several themes from the structured study while introducing one substantively new finding. On workflow features, they independently raised the export/save/share requirement reported by three Group A participants, framing it as a baseline expectation: in a regulated public-sector context, analytical outputs that cannot be saved or shared have limited operational use. Regarding the content, they identified capital and operating expenditure as their primary points of stakeholder interest, consistent with the form-study finding that the headline KPI cards were the most-selected component for communicating the risk comparison.

The new finding concerned traceability of input assumptions. The practitioner observed that the tool currently presents parameter values without documenting their sources or the reasoning behind them, which limits its usefulness in a context where decisions must be defended at a later date or audited by external reviewers. In public-sector water procurement, project authorisation requires written justification of the assumptions underlying the cost estimate. Future iterations should therefore include either a per-input source-citation layer or an exportable methodology summary alongside the numerical outputs.

The practitioner also highlighted the DST’s value as a consultancy tool. Public water utilities and statutory boards routinely commission consultants to produce investment estimates, and the tool’s structured presentation of assumptions, uncertainty, and recommendation was identified as a medium through which consultants could communicate analytical reasoning to non-technical clients. This extends the specialist audience identified in Section 5.5.1: alongside the analyst who interprets and adjusts the analysis, the tool may also serve as the medium through which that analyst communicates results to their client.

This single practitioner perspective does not generalise, but it usefully extends the formative evaluation in two directions. It confirms that the workflow and content priorities identified by specialist participants in the structured study correspond to genuine professional needs, and it surfaces a documentation and traceability requirement that the form-based protocol could not have elicited.

5.6 Summary of findings

The evaluation indicates that the tool achieves its primary objective of supporting specialist analytical work: specialist participants engaged successfully with the distributional outputs and reported the tool as a credible working artefact. The accessibility extension into the non-specialist audience is partial: non-specialists found the interface operable but did not fully understand the technical content underlying the recommendation. Across both

groups, the headline KPI cards emerged as the dominant comprehension anchor, with the more sophisticated visualisations preferred only by specialists or read in combination. The practitioner interview confirmed the workflow priorities surfaced in the structured study and introduced a substantive new requirement: traceability of input assumptions through documented source citations, necessary for the tool to function as a defensible artefact in regulated public-sector contexts. Together these findings indicate that further development should prioritise the content layer (glossary, expanded tooltips, source documentation) and workflow features (export, save, share, presets) over additional analytical capability.

6 Results & Discussion

6.1 NPV and Financial Comparison

The two designs differ structurally in the timing of their capital commitments. The rigid design commits its full €89 M expansion budget across years 0–2 in the predetermined 25%/35%/40% schedule, reaching its nameplate 120 hm³/yr capacity in year 3 regardless of realised demand. The flexible design commits no capital until the rolling utilisation trigger fires, after which each €3.0 M module is added with a one-year build lag. Figure 9 illustrates the annual cash-flow profiles of both designs under the base-case demand realisation.

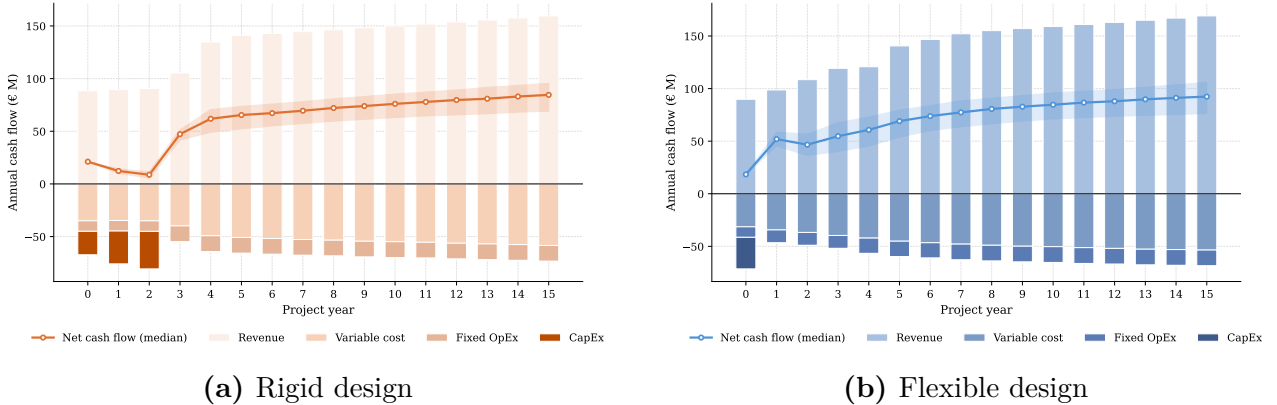


Figure 9: Annual cash-flow profiles of both designs under the base-case demand realisation.

The structural contrast between the two profiles is the mechanism that produces a positive Expected Value of Flexibility. The rigid design commits its capital against a forecast made in year 0 and cannot revise that commitment as the demand trajectory unfolds; the flexible design commits incrementally and matches its capital deployment to realised demand. The flex design’s modules-built trajectory under the base-case realisation is shown in Figure 10: modules are committed incrementally as the rolling utilisation trigger fires, reaching the maximum 22-module envelope progressively across the project horizon.

Aggregated across the 2,000-trial Monte Carlo ensemble, the flexible design dominates the rigid design in expectation. Table 9 reports the headline financial metrics for both designs.

The Expected Value of Flexibility is:

$$\text{EVof} = \text{ENPV}_{\text{flex}} - \text{ENPV}_{\text{rigid}} = \text{€}118.3 \text{ M},$$

and the flexible design outperforms the rigid design in 80.6% of the 2,000 trials. The expected-value advantage is therefore not driven by a small number of extreme outcomes but reflects a consistent structural ordering of the two designs across the simulated futures. Conditional on the flexible design winning, the mean advantage is €166.3 M (1,613 trials);

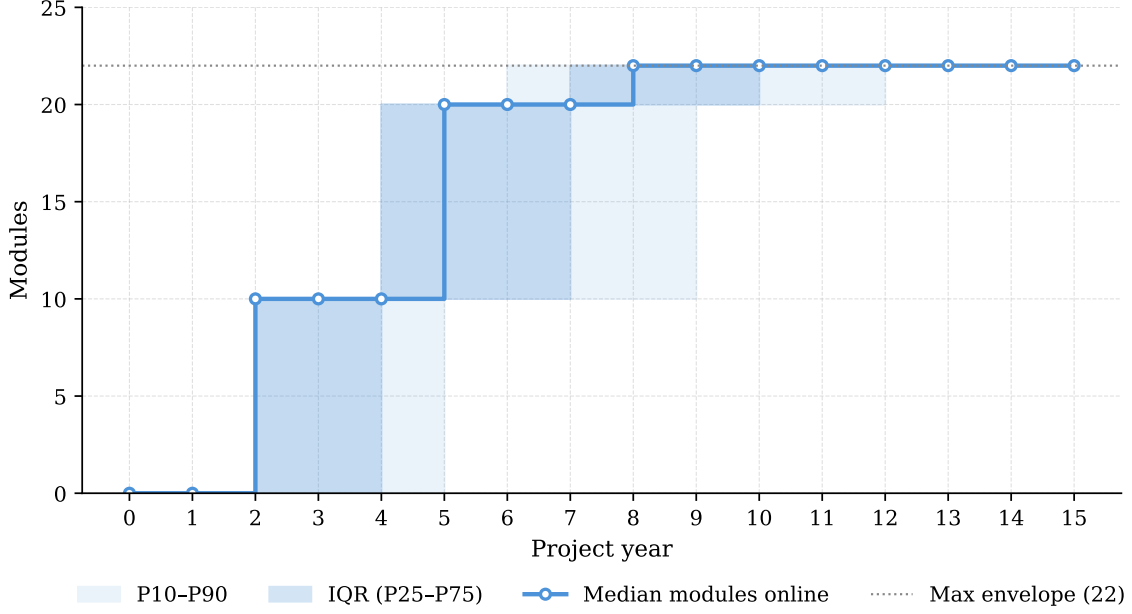


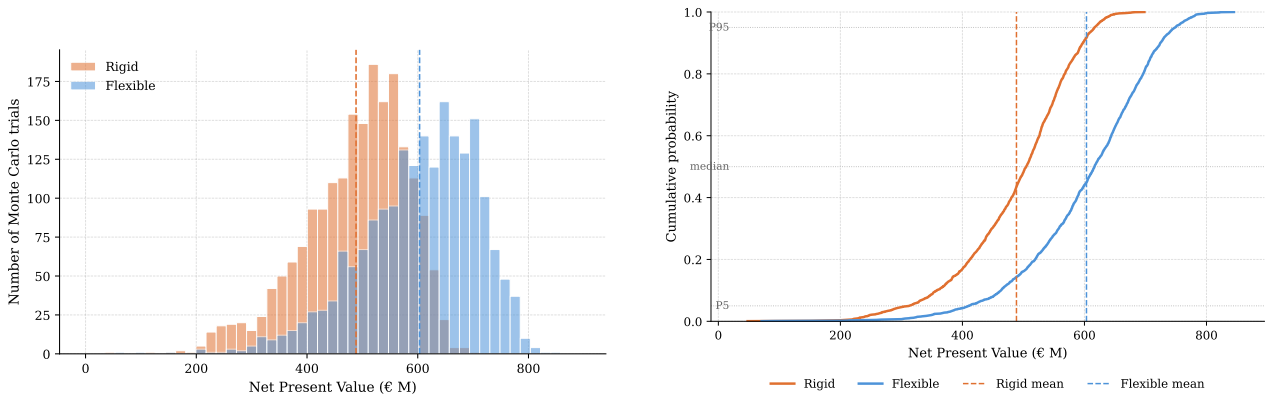
Figure 10: Cumulative modules online for the flexible design under the base-case demand realisation.

Table 9: Financial comparison between the rigid and flexible designs under the active calibration, across $N = 2,000$ Monte Carlo trials.

Metric (€M)	Rigid	Flexible	Flex – Rigid
Mean NPV (ENPV)	487.1	605.4	118.3
Median NPV	499.6	613.4	117.6
Standard deviation	93.2	105.0	—

conditional on the rigid design winning, the mean advantage is €81.8 M (387 trials). The conditional upside is therefore approximately 2.0 times the conditional downside, indicating an asymmetric payoff structure characteristic of an option-bearing design: the flexible plant wins the trials where flexibility matters by a wider margin than the rigid plant wins the trials where it happens to align with demand by chance.

The full distributions of NPV are shown as overlapping histograms in Figure 11a and as cumulative distribution functions in Figure 11b.



(a) Distributions of NPV across the 2,000-trial Monte Carlo ensemble.

(b) Cumulative distribution function (CDF) of NPV.

Figure 11: Rigid and Flexible Net Present Values (NPV)

The CDF separation widens with the percentile level: the flexible curve sits to the right of the rigid curve at every probability level, with a horizontal gap of approximately €113 M at the P5 percentile rising to €142 M at the P95 percentile. The widening gap reflects the asymmetric payoff structure of the embedded expansion option, which captures upside in high-demand realisations while truncating the downside in low-demand realisations through the abandonment mechanism. The distributional structure of this gap is discussed in the next section.

6.2 Distributional Risk and Per-trial Behaviour

The expected-value finding of Section 6.1 does not characterise the risk profile of either design. A higher mean can co-exist with a wider distribution; a design with attractive expected returns may still produce unacceptable downside outcomes in a non-trivial fraction of futures.

Table 10 reports the percentile and spread metrics for each design across the 2,000-trial ensemble.

Table 10: Distributional metrics for each design across $N = 2,000$ Monte Carlo trials. The downside floor and upside ceiling are reported as the 5th and 95th percentiles of NPV.

Metric (€M)	Rigid	Flexible	Flex – Rigid
P5 (downside floor)	310.7	424.0	113.3
Median	499.6	613.4	113.8
P95 (upside ceiling)	613.4	755.2	141.8
P95–P5 spread	302.7	331.1	28.4
Standard deviation	93.2	105.0	—

The flexible design produces a higher downside floor (€424 M vs €311 M at the P5 percentile, a €113 M improvement), a higher upside ceiling (€755 M vs €613 M at the P95 percentile, a €142 M improvement), and a higher median (€613 M vs €500 M). It does so with a marginally wider absolute spread: the P95–P5 range is €331 M for the flexible design against €303 M for the rigid design, and the standard deviation is correspondingly higher (€105 M vs €93 M). The flexible distribution is therefore not narrower in absolute terms. It is a translation of the entire distribution to the right, including the lower tail, accompanied by a modest widening. For a risk-averse decision-maker concerned with bad-case outcomes, this is the relevant comparison: the flexible design’s worst-case outcomes are substantially better, even though its dispersion is larger. The two views of risk — floor versus spread — pull in opposite directions, and the multi-criteria layer in Section 6.4 makes the trade-off explicit.

The per-trial dominance structure of the two designs is summarised in Table 11. In 1,613 of the 2,000 trials (80.6%) the flexible design produces a higher NPV than the rigid design; in the remaining 387 trials (19.4%) the rigid design produces a higher NPV. The conditional mean advantage when the flexible design wins is €166 M; when the rigid design wins it is €82 M — the asymmetric payoff structure introduced in Section 6.1.

Table 11: Per-trial dominance and conditional payoff asymmetry across the 2,000-trial ensemble.

Outcome	Trials	Mean advantage (€M)
Flexible wins (Flex > Rigid)	1,613 (80.6%)	166.3
Rigid wins (Rigid > Flex)	387 (19.4%)	81.8
Asymmetry ratio (upside / downside)	—	2.03

The 2.03 ratio is a direct consequence of the option structure of the flexible design. When realised demand requires capacity expansion, the flexible design captures the upside by exercising its build option. However, when realised demand does not require expansion, the flexible design avoids the stranded capital that suppresses the rigid design's NPV by abandoning. The trials in which the rigid design wins are those in which the pre-committed rigid schedule happens to align with realised demand by chance, but the magnitude of that alignment advantage is limited by the rigid design's lower operational efficiency.

The structural advantage of the flexible design is therefore not a uniform shift but a state-contingent one: the flexible design wins by a wide margin in trials where flexibility matters, and concedes a smaller margin in trials where it does not. This is the canonical signature of an option-bearing design and the empirical justification for the Expected Value of Flexibility framework: a static comparison of mean NPVs understates the true value differential because it averages over a payoff structure that is not symmetric across the realisation space.

6.3 Sensitivity analysis

The headline EVoF of €118 M reported in Section 6.1 is conditional on the deterministic parameter values held fixed during the stochastic simulation. This section interrogates the robustness of the verdict to $\pm 20\%$ perturbations of these parameters.

6.3.1 Deterministic parameter sensitivity

The sensitivity of each design's mean NPV to a $\pm 20\%$ perturbation of each input parameter is reported in Figure 12. The horizontal axis reports the change in mean NPV from the base case, with parameters ordered by absolute swing magnitude. Each design is perturbed independently in its own ensemble, so the two tornados are read as parallel descriptions of how each design responds to parameter variation rather than as a strict EVoF sensitivity.

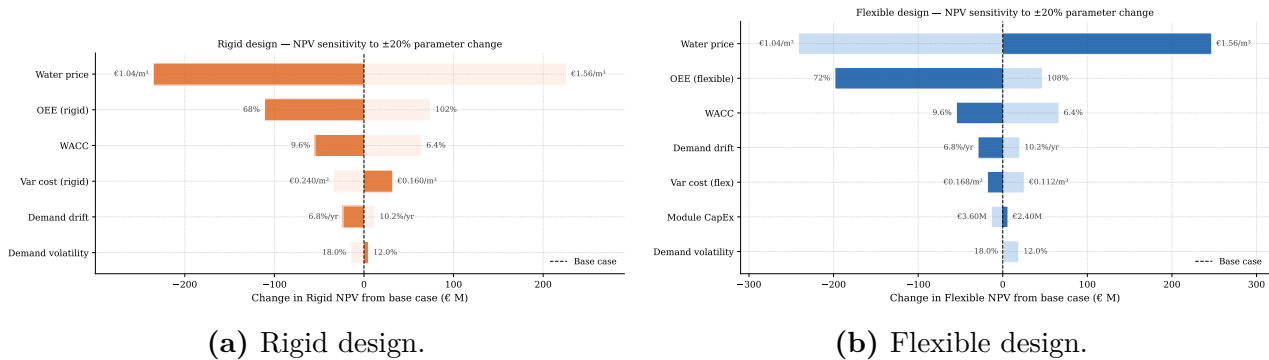


Figure 12: Tornado sensitivity of mean NPV to a $\pm 20\%$ perturbation of each input parameter for both designs.

Three findings follow from the comparison.

First, water price dominates the sensitivity of both designs. A $\pm 20\%$ perturbation moves both designs' mean NPV by hundreds of millions of euros, with the two swings nearly symmetric within each design (lower price reduces NPV by the same magnitude that higher price increases it) and comparable in direction across the two designs. Because both designs respond to water price by similar magnitudes in the same direction, the EVoF is largely preserved across this perturbation: the parameter that most affects absolute NPV is not the parameter that most affects the difference between the designs.

Second, the operational efficiency parameter exhibits an asymmetric response in both designs.

Improvements above the base case deliver limited additional value, but degradations from it have substantial cost; this is the canonical asymmetry of a binding capacity ceiling against an open downside. The flexible design’s nominal 90% OEE is close to a practical ceiling. Operational performance shortfalls are therefore the most consequential single-parameter risk to either design’s expected return.

Third, the module CapEx — the parameter unique to the flexible design — has a comparatively small effect on flexible NPV. A $\pm 20\%$ perturbation of the per-module cost moves the flexible mean NPV by a margin much smaller than that produced by either water price or OEE. Within the surrounding scenarios of the base case, the flexible design’s expected value is relatively insensitive to module cost.

The discount rate, variable cost, demand drift, and demand volatility are second-tier influences in both tornados. Demand volatility is the smallest contributor to either design’s NPV in the one-at-a-time analysis, consistent with the framework’s location of volatility’s principal effect in the option-value calculation rather than in the expected-value calculation.

6.4 Multi-criteria perspective

The financial and distributional findings of Sections 6.1 and 6.2 evaluate both designs on a single axis of value. The multi-criteria layer specified in Section 3.5 broadens this comparison across four axes — Economic, Environmental, Operational, and Predictability — each scored on a 0–100 scale from the Monte Carlo ensemble. Table 12 reports the per-axis scores and composite weighted score for each design under equal axis weighting.

Table 12: Multi-criteria scores for the rigid and flexible designs on a 0–100 scale under equal axis weighting.

Criterion	Rigid	Flexible	Basis
Economic	15.0	100.0	Normalised mean NPV (floor-based scaling)
Environmental	15.0	100.0	Mean fraction of demand met
Operational	57.5	78.9	50% utilisation + 50% CapEx schedule certainty
Predictability	100.0	15.0	Inverse of P95–P5 NPV spread
Composite (equal weights)	46.9	73.5	

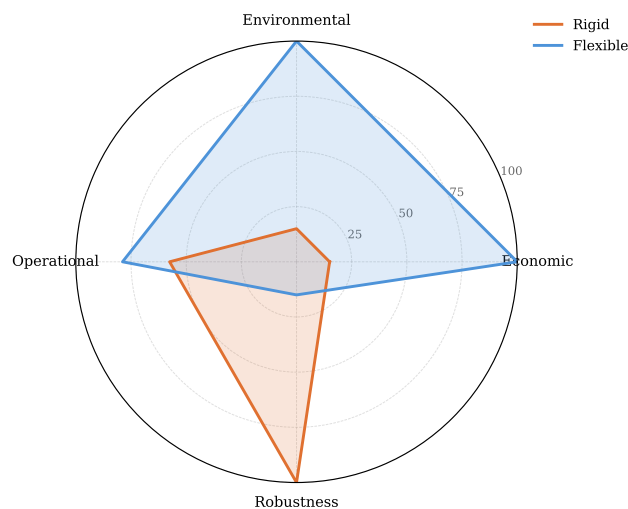


Figure 13: Multi-criteria scores by axis for the rigid and flexible designs.

The flexible design leads on three of the four axes. The Economic advantage reflects the €118 M mean NPV differential reported in Section 6.1: under floor-based scaling against the joint range of both designs, the flexible design’s higher mean translates directly into the maximum economic score. The Environmental advantage reflects its consistently higher fraction of realised demand met across the ensemble, a consequence of the modular design’s ability to add capacity in response to demand growth rather than relying on a pre-committed nameplate envelope. The Operational advantage is consistent, with the flexible design’s higher utilisation outweighing the rigid design’s CapEx-schedule certainty in the composite definition.

The Predictability axis, by contrast, favours the rigid design. The axis is operationalised through the inverse of the P95–P5 NPV spread: a tighter distribution corresponds to a more predictable outcome and a higher score. This is a deliberately narrow operationalisation of robustness as outcome *predictability*; an alternative operationalisation as downside protection (the P5 floor) would reverse the verdict, since the flexible design’s P5 of €424 M substantially exceeds the rigid design’s P5 of €311 M (Section 6.2). The axis label has been chosen to make this scope explicit: a stakeholder who cares about worst-case outcomes should down-weight Predictability and consult the P5 floor directly. The rigid design’s deterministic CapEx schedule and fixed operating profile produce a marginally tighter NPV distribution (P95–P5 spread of €303 M) than the flexible design’s demand-tracking expansion path (€331 M), and the axis scoring reflects this.

Under equal axis weighting the composite score is 73.5 for the flexible design against 46.9 for the rigid design, a margin sufficiently large that the verdict is preserved across most plausible weighting combinations. A weight-sensitivity check confirms that the rigid design achieves a higher composite only when the Predictability axis is weighted above approximately 55%, a weighting that few public-sector decision-makers would adopt in isolation. The multi-criteria perspective therefore reinforces, rather than contradicts, the financial finding: the flexible design dominates on three substantive performance dimensions and concedes only the dispersion axis, in which the trade-off is between predictability and expected value rather than between performance and underperformance.

Taken together, the findings of this section support a consistent conclusion across the analytical dimensions reported. The flexible design produces a higher expected NPV, a substantially better downside floor, an EVoF that remains positive across $\pm 20\%$ deterministic parameter perturbations, and higher composite scores on three of four multi-criteria axes. The single dimension on which the rigid design leads — predictability as defined by NPV dispersion — reflects a structural property of the comparison rather than a performance shortfall of the flexible design, and is unlikely to outweigh the economic, environmental, and operational advantages under any defensible weighting scheme.

7 Conclusion

7.1 Overview and contributions

This thesis developed an integrated framework for valuing flexibility in modular desalination plant investments under demand and energy-cost uncertainty, combining real-options analysis, robustness assessment, and multi-criteria decision analysis, delivered through an interactive Decision-Support Tool. The framework was applied to the Torrevieja SWRO plant as a case study comparing a rigid staged expansion against a modular incremental alternative across a 15-year horizon.

The thesis makes four contributions. The first is a unified framework that integrates Real

Options Analysis, Monte Carlo robustness assessment, and multi-criteria decision analysis within a single computational protocol. Real-options, robustness, and MCDA layers are typically applied in isolation in the water-infrastructure literature; here they are combined through paired Monte Carlo sampling and a deliberately competitive MCDA specification that avoids double-counting the economic axis already captured by the real-options layer.

The second is a bottom-up techno-economic model of SWRO desalination capacity expansion. The model is calibrated against published Spanish coastal water-demand and Iberian electricity-market data, encodes an explicit modular-expansion trigger rule with build-lag and abandonment behaviour, and differentiates the cash-flow architecture of the rigid and flexible designs through their respective operational efficiencies and CapEx schedules.

The third is a decision-support artefact: an interactive web-based tool designed primarily for specialist users while extending accessibility to non-specialist stakeholders, validated through a formative usability evaluation with ten participants and a complementary practitioner interview.

The fourth is an empirical assessment of when modular flexibility is valuable in SWRO investment. Applied to Torrevieja, the framework produces an Expected Value of Flexibility of €118 M under the active calibration, with the flexible design outperforming the rigid design in 81% of 2,000 Monte Carlo trials. The verdict is robust across $\pm 20\%$ deterministic parameter perturbations. The multi-criteria layer shows the flexible design leading on the economic, environmental, and operational axes, while the rigid design leads on the predictability axis as defined by NPV dispersion — a finding that highlights the difference between robustness understood as a tighter distribution and robustness understood as a higher downside floor, and motivates the explicit weight-sensitivity exposure in the Decision-Support Tool.

Taken together, the contributions extend the existing literature on flexibility valuation in water infrastructure by integrating it with robust decision-making approaches and by delivering the analytical machinery in a form designed for direct stakeholder use. The framework is not specific to desalination: any infrastructure investment with sequential capacity-expansion options and stochastic demand and cost drivers can be analysed within the same structure.

7.2 Study limitations

Seven limitations frame the interpretation of the findings. The stochastic process specification represents demand and electricity price as independent geometric Brownian motions, imposing log-normality on the distribution of future states and excluding mean-reverting dynamics, jump processes, and cross-correlation between the two drivers. The decision rule governing the flexible design’s capacity-expansion decisions is a threshold heuristic rather than a fully optimal dynamic programme, so the reported EVoF should be interpreted as a lower bound on the true option value. The empirical evidence is drawn from a single case study; the findings are specific to Torrevieja’s operational parameters and have not been replicated across multiple installations.

The cost-architecture assumption sets the per-module CapEx at €3.0 M, reflecting modular SWRO train pricing and yielding a flexible total expansion outlay below that of the rigid build. This biases the comparison toward the modular design and is documented as a structural choice; project-specific cost estimates would be required before the framework’s quantitative recommendations could be applied to a real procurement decision. Conversely and relatedly, the analysis assigns the flexible design a 5-percentage-point OEE advantage and a 30% non-energy variable-cost advantage relative to the rigid design, reflecting the newer-vintage membranes, energy-recovery devices, and modular-maintenance scheduling expected on staged-commissioning capacity. These vintage advantages are real but they are

conceptually distinct from the option-timing value of flexibility; the reported EVoF is therefore the combined advantage of the modular architecture rather than a pure option premium. A decomposition simulation holding OEE and variable cost equal across designs — isolating the option-timing component — is identified as the highest-priority extension in Section 7.3.

The usability evaluation involved ten participants, all from Engineering, Mathematics, or Computer Science backgrounds, and the participants assumed the role of a junior analyst rather than being practising water-infrastructure planners; the evaluation produces design signals for an early-stage artefact rather than confirmatory validation. The protocol used ad-hoc rating scales rather than validated psychometric instruments (SUS, NASA-TLX, UEQ); a summative evaluation appropriate for confirmatory claims would replace these with the validated instruments, as flagged in Section 5.5.1. Finally, the multi-criteria layer uses axis definitions, scoring methodology, and weights specified within the thesis’s methodological frame rather than elicited from stakeholders, a participatory step that would strengthen the framework’s claim to represent the criteria that genuinely matter to public-sector decision-makers.

7.3 Further work

Five directions for further work emerge from the limitations above. The first is a structural decomposition of the Expected Value of Flexibility into its option-timing and technology-vintage components. The present framework reports the combined advantage of the modular design; an extension holding OEE and non-energy variable cost equal across the two designs would isolate the pure option-timing premium, and a regime sweep across the (demand drift, demand volatility) plane at equal vintage assumptions would identify the conditions under which the option-timing component dominates — directly delivering Objective 4 in a more discriminating form than the present base-case finding.

Methodologically, the framework can also be extended to mean-reverting and jump-augmented stochastic processes, to bivariate demand-energy models that capture the correlation the present specification treats as zero, and to Least-Squares Monte Carlo decision rules. A fully optimal LSMC policy would produce an upper bound on the Expected Value of Flexibility that, bracketed against the present heuristic lower bound, would tighten the framework’s quantitative claims.

Empirically, the framework should be applied to additional SWRO contexts to test the generalisability of the case-study findings. Ex-post validation against historical investment decisions — comparing the framework’s recommendations to actual modular SWRO deployments and their realised outcomes — would provide stronger empirical grounding than any single forward-looking case study can.

The Decision-Support Tool admits both incremental improvements and a more substantive product direction. The priority incremental improvements identified by the usability evaluation and practitioner interview concern the content layer: a dedicated glossary, expanded hover-to-define coverage for acronyms and technical terms, and source-citation traceability for every input parameter to support public-sector audit and project-authorisation contexts. Workflow features — export, save, share, parameter presets, and simplified-versus-advanced modes — were independently requested as baseline requirements for professional use. A summative usability evaluation applying SUS, NASA-TLX, and UEQ on the next iteration would convert the present formative signals into confirmatory claims about decision-support efficacy. Beyond these, the practitioner interview surfaced a substantively new direction: a consultancy-mode export delivering the assumption-uncertainty-recommendation triad as a structured client artefact, positioning the tool not only for analytical work but as a medium of transparent technical communication between consultants and their public-sector clients.

Finally, a participatory specification of the multi-criteria layer through structured workshops with water-utility planners, regulators, and community representatives would replace the methodologically-specified axis weights and scoring with a stakeholder-elicited version, strengthening the framework’s claim to represent the criteria that genuinely matter to those who must act on its recommendations.

7.4 Professional reflection

This project was an incredible opportunity for both technical and personal growth. It aligned with my interests in strategic infrastructure investment and the challenge of making technical analyses accessible to those who must act on them, while my background in economics and finance, systems design, and programming supported the development of the simulation engine and the Decision-Support Tool.

A key area of growth was the bridging of analytical depth with stakeholder communication. Building the Monte Carlo and Real Options layers was a familiar technical exercise; designing how those outputs should be presented to non-specialist users was not. Supervisor feedback on the multi-criteria specification — particularly on avoiding double-counting and making the dispersion-versus-downside trade-off explicit — pushed me toward a more deliberate analytical structure. The usability sessions later imposed a similar discipline on the interface: every term participants asked me to explain, and every chart that produced hesitation, became a candidate for redesign.

Benchmarked against directly comparable Master’s-level work, the project sits closest in scope to Bigestans (2023) on flexible blue-hydrogen production and Najafi & Aghaei (2024) on photovoltaic-powered RO, both of which apply ROA to capacity-expansion problems but neither of which delivers a stakeholder-facing decision tool or an integrated four-axis MCDA layer. The DST is the principal distinguishing contribution; the technology-vintage decomposition is the principal area where a published version of this work would extend the present analysis.

The evaluation also revealed a tendency to optimise the analytical machinery before the communication layer: non-specialists engaged most readily with the simplest visual elements rather than the analytical sophistication invested in the CDF and tornado views. Going forward I aim to treat technical refinement and communication design as parallel rather than sequential workstreams.

Overall, the project tested both technical and professional maturity, requiring interdisciplinary thinking across stochastic modelling, infrastructure economics, and human-factors design — foundations I intend to build on as I move into industry.

7.5 Project Timeline and Workflow

A structured and adaptive project-management approach was adopted across the two academic terms. The initial plan, set out in the Early Stage Gateway literature review, sequenced six objectives — Literature Review, Framework Development, MCDA Specification, Flexibility Analysis, Stress Testing under Uncertainty, and Final Report — organised across four operational phases: Foundation & Planning, Model Development, Testing & Evaluation, and Reporting.

The original plan was followed closely through the first three phases. Foundation delivered the literature review, methodological framework, and case-study scoping. Model Development built the techno-economic model in Excel, implemented the modular-expansion trigger logic, and calibrated the stochastic processes against historical Spanish data. Testing ran the paired Monte Carlo ensemble, conducted the sensitivity analyses, and assembled the multi-criteria

scoring layer.

Following the Late Stage Gateway, a revised timeline was introduced to accommodate a substantively new deliverable: a usability-evaluated Decision-Support Tool. This required re-implementing the model in Python (Plotly Dash), designing the five-tab interface, running a ten-participant usability study with a complementary practitioner interview, and integrating the findings into Section 5.

The DST insertion compressed the Reporting phase and required tasks to be sequenced by critical dependency, with non-critical work deferred to the final two weeks. The trade-off was an additional contribution and a stronger evaluation chapter at the cost of a tighter reporting window — a sequencing choice I would make again given the same constraints.

8 Acknowledgements

I am incredibly grateful to my supervisor, Dr. Michel-Alexandre Cardin, for his invaluable guidance and mentorship throughout this project. I am also thankful to the faculty and staff at the Dyson School of Design Engineering for providing the resources and academic environment necessary to complete this work. Finally, thank you to my family for their unwavering love and support throughout all my academic journey.

I acknowledge the use of Claude (Anthropic, <https://claude.ai>) in the development of this report. Specifically, the tool was used to assist with the technical implementation of the Python code, to refine the structural formatting of tables and figures, and to improve the conciseness of the text. I confirm that all analysis, findings, and conclusions presented in this report are my own work.

References

- [1] E. Jones, M. Qadir, M. T. van Vliet, V. Smakhtin, and S. mu Kang, “The state of desalination and brine production: A global outlook,” *Science of The Total Environment*, vol. 657, pp. 1343–1356, 2019, ISSN: 0048-9697. DOI: <https://doi.org/10.1016/j.scitotenv.2018.12.076>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0048969718349167>.
- [2] MedECC. “The state of desalination and brine production: A global outlook.” Accessed: 2026-06-02. [Online]. Available: <https://www.medecc.org/the-state-of-desalination-and-brine-production-a-global-outlook-article/>.
- [3] R. J. Lempert and D. G. Groves, “Identifying and evaluating robust adaptive policy responses to climate change for water management agencies in the american west,” *Technological Forecasting and Social Change*, vol. 77, no. 6, pp. 960–974, 2010, Two Special Sections: Risk and Technology Addressing Deep Uncertainty Using Adaptive Policies, ISSN: 0040-1625. DOI: <https://doi.org/10.1016/j.techfore.2010.04.007>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0040162510000740>.
- [4] E. Borgomeo, M. Mortazavi-naeini, J. Hall, and B. Guillod, “Risk, robustness and water resources management under uncertainty,” *Earth’s Future*, vol. 6, Feb. 2018. DOI: [10.1002/2017EF000730](https://doi.org/10.1002/2017EF000730). [Online]. Available: https://www.researchgate.net/publication/323154653_Risk_Robustness_and_Water_Resources_Management_under_Uncertainty.
- [5] D. G. G.-N. K. J. R. F. S. C. B. B. P. B. M. T. C. K. K. A. H. e. a. Robert J. Lempert Steven W. Popper, “Making good decisions without predictions,” *RAND*, 2013, Accessed: 2025-06-10. [Online]. Available: https://www.rand.org/pubs/research_briefs/RB9701.html.
- [6] M. Haasnoot, J. H. Kwakkel, W. E. Walker, and J. ter Maat, “Dynamic adaptive policy pathways: A method for crafting robust decisions for a deeply uncertain world,” *Global Environmental Change*, 2013. DOI: <https://doi.org/10.1016/j.gloenvcha.2012.12.006>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S095937801200146X>.
- [7] S. R. Downward and R. Taylor, “An assessment of spain’s programa agua and its implications for sustainable water management in the province of almería, southeast spain,” *Journal of Environmental Management*, vol. 82, no. 2, pp. 277–289, 2007, ISSN: 0301-4797. DOI: <https://doi.org/10.1016/j.jenvman.2005.12.015>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301479706000375>.
- [8] K. Pachos, I. Huskova, E. Matrosov, T. Erfani, and J. J. Harou, “Trade-off informed adaptive and robust real options water resources planning,” *Advances in Water Resources*, 2022. DOI: <https://doi.org/10.1016/j.advwatres.2021.104117>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0309170821002669>.
- [9] S. Fletcher, M. Lickley, and K. Strzepek, “Learning about climate change uncertainty enables flexible water infrastructure planning,” *Nature Communications*, 2019. DOI: [10.1038/s41467-019-10411-7](https://doi.org/10.1038/s41467-019-10411-7).

- 1038/s41467-019-09677-x. [Online]. Available: https://www.researchgate.net/publication/332457355_Learning_about_climate_change_uncertainty_enables_flexible_water_infrastructure_planning.
- [10] R. de Neufville and S. Scholtes, *Flexibility in engineering design*, Accessed: 2025-06-10, 2011. [Online]. Available: <https://direct.mit.edu/books/monograph/2955/Flexibility-in-Engineering-Design>.
 - [11] P. Suttinon and S. Nasu, "Real options for increasing value in industrial water infrastructure," *Water Resources Management*, 2010. DOI: 10.1007/s11269-010-9585-0. [Online]. Available: https://www.researchgate.net/publication/225524491_Real_Options_for_Increasing_Value_in_Industrial_Water_Infrastructure.
 - [12] M. Mahdi, S. Xueqian, Q. Gai, M. Basirialmahjough, and H. Yuan, "Improving robustness of water supply system using a multi-objective robust optimization framework," *Environmental Research*, 2023. DOI: <https://doi.org/10.1016/j.envres.2023.116270>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0013935123010745>.
 - [13] L. Pearson, A. Coggan, W. Proctor, and T. Smith, "A sustainable decision support framework for urban water management," *Water Resources Management*, 2009. DOI: 10.1007/s11269-009-9450-1. [Online]. Available: https://www.researchgate.net/publication/226061563_A_Sustainable_Decision_Support_Framework_for_Urban_Water_Management.
 - [14] E. Kuznetsova, M.-A. Cardin, M. Diao, and S. Zhang, "Integrated decision-support methodology for combined centralized-decentralized waste-to-energy management systems design," *Renewable and Sustainable Energy Reviews*, 2019. DOI: <https://doi.org/10.1016/j.rser.2018.12.020>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364032118308165>.
 - [15] F. Black and M. Scholes, "The pricing of options and corporate liabilities," *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973, ISSN: 00223808, 1537534X. Accessed: Jun. 4, 2026. [Online]. Available: <http://www.jstor.org/stable/1831029>.
 - [16] J. Marques, M. Cunha, and D. Savic, "Using real options in the optimal design of water distribution networks," *Journal of Water Resources Planning and Management*, 2015. DOI: 10.1061/(ASCE)WR.1943-5452.0000448. [Online]. Available: https://www.researchgate.net/publication/271501128_Using_Real_Options_in_the_Optimal_Design_of_Water_Distribution_Networks.
 - [17] S. Zhang and V. Babovic, "A real options approach to the design and architecture of water supply systems using innovative water technologies under uncertainty," *Journal of Hydroinformatics*, 2012. DOI: 10.2166/hydro.2011.078. [Online]. Available: https://www.researchgate.net/publication/249643295_A_Real_Options_Approach_to_the_Design_and_Architecture_of_Water_Supply_Systems_Using_Innovative_Water_Technologies_Under_Uncertainty.
 - [18] Y. Deng, M.-A. Cardin, V. Babovic, D. Santhanakrishnan, P. Schmitter, and A. Meshgi, "Valuing flexibilities in the design of urban water management systems," *Water Research*, 2013. DOI: <https://doi.org/10.1016/j.watres.2013.09.064>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0043135413008312>.
 - [19] Y. Wang, M. Zechner, J. M. Mern, M. J. Kochenderfer, and J. K. Caers, "A sequential decision-making framework with uncertainty quantification for groundwater management," *Advances in Water Resources*, vol. 166, p. 104266, 2022, ISSN: 0309-1708. DOI: <https://doi.org/10.1016/j.adwatres.2022.104266>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0309170822001324>.
 - [20] A. Hurford et al., "Efficient and robust hydropower system design under uncertainty - a demonstration in nepal," *Renewable and Sustainable Energy Reviews*, 2020, ISSN:

- 1364-0321. DOI: <https://doi.org/10.1016/j.rser.2020.109910>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S136403212030201X>.
- [21] M. Brugnach, I. Ruiz, N. Zafra-Calvo, L. Vivas, and M.-J. Sanz-Sanchez, "Advancing decision-making amid uncertainty in water governance and management under climate change," *PLOS Climate*, vol. 4, May 2025. DOI: 10.1371/journal.pclm.0000617.
- [22] V. Belton and T. Stewart, "Multiple criteria decision analysis," vol. 142, Jan. 2002. DOI: 10.1007/978-1-4615-1495-4_1.
- [23] B.-H. O. Radmehr Ahmade and H. A. Loáiciga, "Integrated strategic planning and multi-criteria decision-making framework with its application to agricultural water management," *Scientific Reports*, 2022. [Online]. Available: <https://doi.org/10.1038/s41598-022-12194-5>.
- [24] S. Hajkowicz and A. Higgins, "A comparison of multiple criteria analysis techniques for water resource management," *European Journal of Operational Research*, vol. 184, no. 1, pp. 255–265, 2008, ISSN: 0377-2217. DOI: <https://doi.org/10.1016/j.ejor.2006.10.045>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0377221706011039>.
- [25] C. Makropoulos, K. Natsis, S. Liu, K. Mittas, and D. Butler, "Decision support for sustainable option selection in integrated urban water management," *Environmental Modelling Software*, vol. 23, no. 12, pp. 1448–1460, 2008, ISSN: 1364-8152. DOI: <https://doi.org/10.1016/j.envsoft.2008.04.010>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364815208000698>.
- [26] N. H. Afgan, M. Darwish, and M. G. Carvalho, "Sustainability assessment of desalination plants for water production," *Desalination*, vol. 124, no. 1, pp. 19–31, 1999, European Conference on Desalination and the Environment, ISSN: 0011-9164. DOI: [https://doi.org/10.1016/S0011-9164\(99\)00085-5](https://doi.org/10.1016/S0011-9164(99)00085-5). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0011916499000855>.
- [27] T. Comes, M. Hiete, N. Wijngaards, and F. Schultmann, "Decision maps: A framework for multi-criteria decision support under severe uncertainty," *Decision Support Systems*, vol. 52, no. 1, pp. 108–118, 2011, ISSN: 0167-9236. DOI: <https://doi.org/10.1016/j.dss.2011.05.008>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0167923611001163>.
- [28] A. Tsoukiàs, "From decision theory to decision aiding methodology," *European Journal of Operational Research*, vol. 187, no. 1, pp. 138–161, 2008, ISSN: 0377-2217. DOI: <https://doi.org/10.1016/j.ejor.2007.02.039>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0377221707002949>.
- [29] M. Tariq, R. Alotaibi, K. Weththasinghe, and Z. Rajabi, "A detailed perspective of water resource management in a dry and water scarce country: The case in kuwait," *Frontiers in Environmental Science*, vol. 10, p. 1073834, Dec. 2022. DOI: 10.3389/fenvs.2022.1073834.
- [30] G.-P. I. R.-M. J. G.-G. A. Cruz-Pérez N Santamarta JC, "A comparison between carbon footprint of water production facilities in the canary islands: Groundwater resources vs. seawater desalination. sustain water resour manag.," *National Library of Medicine*, 2022. DOI: 10.1007/s40899-022-00706-0.
- [31] N. Ghaffour, T. M. Missimer, and G. L. Amy, "Technical review and evaluation of the economics of water desalination: Current and future challenges for better water supply sustainability," *Desalination*, vol. 309, pp. 197–207, 2013, ISSN: 0011-9164. DOI: <https://doi.org/10.1016/j.desal.2012.10.015>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0011916412005723>.
- [32] Sacyr. "Torrevieja desalination plant expansion." Accessed: 2026-06-01. [Online]. Available: <https://sacyr.com/en/-/ampliacion-desaladora-torrevieja>.

- [33] Aguas de las Cuencas Mediterráneas, *Desaladora de Torrevieja: Infraestructuras de la Acuamed*, Ministerio para la Transición Ecológica y el Reto Demográfico, Accessed: 2026-05-27, 2026. [Online]. Available: <https://www.acuamed.es/project/desaladora-de-torrevieja/>.
- [34] N. Voutchkov, "Energy use for membrane seawater desalination – current status and trends," *Desalination*, vol. 431, pp. 2–14, 2018, "Desalination, energy and the environment" in honor of Professor Raphael Semiat, ISSN: 0011-9164. DOI: <https://doi.org/10.1016/j.desal.2017.10.033>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0011916417321057>.
- [35] Mancomunidad de los Canales del Taibilla, *Demanda de Agua: Datos de Consumo de los Ayuntamientos de la MCT*, Ministerio para la Transición Ecológica y el Reto Demográfico, Accessed: 2026-05-26, 2026. [Online]. Available: https://www.mct.es/web/mct/-/ban_demanda-de-agua.
- [36] Ministerio para la Transición Ecológica y el Reto Demográfico, "Seguridad Hídrica en España: Informe de Gestión y Planificación," Gobierno de España, Official Report, 2023.
- [37] European Commission, *Eu blue economy sectors: Desalination*, EU Blue Economy Observatory, Accessed: 2026-05-26, 2026. [Online]. Available: https://blue-economy-observatory.ec.europa.eu/eu-blue-economy-sectors/desalination_en.
- [38] Operador del Mercado Ibérico de Energía, *Day-ahead minimum, average and maximum price*, OMIE Market Data Portal, Accessed: 2026-05-27, 2026. [Online]. Available: <https://www.omie.es/en/market-results/daily/prices/day-ahead-minimum-average-and-maximum-price>.
- [39] M.-A. Cardin, M. Ranjbar-Bourani, and R. de Neufville, "Improving the lifecycle performance of engineering projects with flexible strategies: Example of on-shore lng production design," *Systems Engineering*, vol. 18, Apr. 2015. DOI: 10.1002/sys.21301.
- [40] D. Bigestans, M.-A. Cardin, and N. Kazantzis, "Economic performance evaluation of flexible centralised and decentralised blue hydrogen production systems design under uncertainty," *Applied Energy*, vol. 352, p. 121944, 2023, ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2023.121944>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0306261923013089>.
- [41] A. Benahmed et al., "Investigating the long-term economic sustainability and water production costs of desalination plants: A case study from chatt hilal in algeria," *Egyptian Journal of Aquatic Research*, vol. 51, no. 1, pp. 31–38, 2025, ISSN: 1687-4285. DOI: <https://doi.org/10.1016/j.ejar.2024.11.011>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1687428524001080>.

9 Appendices

9.1 Nomenclature

Acronyms and abbreviations

Acronym	Definition
CapEx	Capital expenditure
CDF	Cumulative distribution function
CRN	Common random numbers (variance-reduction technique)
DST	Decision-Support Tool
EBITDA	Earnings before interest, taxes, depreciation, and amortisation
ENPV	Expected Net Present Value (across Monte Carlo ensemble)
ERD	Energy Recovery Device
EVof	Expected Value of Flexibility ($\text{ENPV}_{\text{flex}} - \text{ENPV}_{\text{rigid}}$)
GBM	Geometric Brownian Motion
IQR	Interquartile range
KPI	Key Performance Indicator
LSMC	Least-Squares Monte Carlo
MCDA	Multi-Criteria Decision Analysis
NASA-TLX	NASA Task Load Index (validated cognitive-load instrument)
NPV	Net Present Value
OEE	Overall Equipment Effectiveness
OMIE	Operador del Mercado Ibérico de Energía (Iberian electricity market operator)
OpEx	Operating expenditure
P5, P95	5th and 95th percentile of a distribution
ROA	Real Options Analysis
SE	Standard error
SUS	System Usability Scale (validated usability instrument)
SWRO	Seawater Reverse Osmosis
UEQ	User Experience Questionnaire (validated UX instrument)
UI	User Interface
WACC	Weighted Average Cost of Capital

Roman symbols

Symbol	Definition	Units
C_t	Annual cash flow at time t	
D_t	Realised demand at time t	m^3/yr
E_t	Electricity price at time t	$/\text{kWh}$
I_t	Installed capacity at time t	m^3/yr
K	Capital cost (one-off or per-module)	
M_t	Cumulative modules online at time t	—
$M_{\max}$	Maximum module envelope	22
N	Number of Monte Carlo trials	—
P_t	Water tariff at time t	$/\text{m}^3$
Q_t	Production volume at time t	m^3/yr
r	Discount rate (WACC)	—
T	Project planning horizon	15 yr
t	Project year index, $t = 0, 1, \dots, T$	—
u_t	Two-year rolling utilisation at time t	—
Z_t	Standard normal innovation at time t	—

Greek symbols

Symbol	Definition	Units
η	Overall Equipment Effectiveness (OEE)	—
μ	Drift parameter of geometric Brownian motion	yr^{-1}
μ_D	Demand drift	yr^{-1}
μ_E	Electricity-price drift	yr^{-1}
σ	Volatility parameter of geometric Brownian motion	yr^{-1}
σ_D	Demand volatility	yr^{-1}
σ_E	Electricity-price volatility	yr^{-1}
τ_{build}	Build trigger threshold (rolling utilisation)	—
τ_{abandon}	Abandonment trigger threshold (rolling utilisation)	—

Subscripts

Subscript	Meaning
D	Demand process
E	Electricity-price process
flex	Flexible (modular) design
rigid	Rigid (staged) design
t	Time index
0	Project commencement (year 0)

9.2 Stochastic Calibration: Source Data

This appendix lists the raw time-series data and the literature sources underlying the calibrations reported in Section 4.3. The fitting methodology and the rationale for the active mix-and-match selection are presented in the main text and are not repeated here.

9.2.1 Water demand

Annual desalinated water demand figures for the Vega Baja region were obtained from the Mancomunidad de los Canales del Taibilla (MCT) annual reports [35] and the regional water-security monitoring publication [36]. Raw values are tabulated in Table 13.

Year	Demand (hm ³ /yr)	Year	Demand (hm ³ /yr)
2003	5.01	2015	38.56
2004	16.82	2016	59.51
2005	21.65	2017	85.37
2006	42.01	2018	92.87
2007	60.25	2019	68.49
2008	72.36	2020	72.45
2009	72.61	2021	64.24
2010	56.20	2022	80.72
2011	34.87	2023	74.40
2012	44.03	2024	100.80
2013	6.27	2025	69.10
2014	11.23		

Table 13: Annual desalinated water demand for the Vega Baja region 2003–2025. Source: MCT annual reports and Seguridad Hídrica monitoring publications.

The literature-based drift of $\mu_D = 0.07/\text{yr}$ is taken from European Commission long-term demand projections for Mediterranean water-stressed regions [37]. The literature volatility of $\sigma_D = 0.15/\text{yr}$ is consistent with comparable real-options studies ([39]: 5–12%/yr; [40]: 8–15%/yr).

9.2.2 Electricity price

Annual average wholesale electricity prices for the Iberian market 2010–2025 were obtained from OMIE [38] and the Datos.macro aggregated commodity index. Raw values are tabulated in Table 14.

Year	Price (€/kWh)	Year	Price (€/kWh)
2010	0.0370	2018	0.0573
2011	0.0499	2019	0.0477
2012	0.0472	2020	0.0339
2013	0.0443	2021	0.1114
2014	0.0421	2022	0.1675
2015	0.0503	2023	0.0871
2016	0.0397	2024	0.0630
2017	0.0522	2025	0.0700

Table 14: Annual average wholesale electricity prices, Iberian market 2010–2025. Source: OMIE day-ahead annual averages.

The literature volatility of $\sigma_E = 0.12/\text{yr}$ reflects pre-crisis Iberian wholesale price volatility and the conservative central estimates adopted by [39] and [40] for energy-cost uncertainty in long-horizon real-options studies.

9.3 Decision-Rule and Operational Parameter Sources

Each decision-rule and operating parameter introduced in Section 4.4 is sourced as documented in Table 15. The table is provided to support reproducibility and to expose the empirical basis for each value held fixed in the stochastic simulation.

Table 15: Source and rationale for decision-rule and operational parameters.

Parameter	Value	Source
<i>Plant configuration</i>		
Baseline capacity	80 hm ³ /yr	Existing Torrevieja nameplate before expansion [32].
Target capacity (post-expansion)	120 hm ³ /yr	Announced expansion: +40 hm ³ /yr added to baseline [32].
Project horizon	15 yrs	2025 (year 0) through 2040 (year 15).
<i>Capital and operating costs</i>		
Total rigid CapEx	€89 M	Announced investment for the +40 hm ³ /yr expansion [32].
CapEx per module (flexible)	€3.0 M	Modular SWRO train procurement at industry pricing.
Fixed OpEx (baseline)	€10 M/yr	Plant-level fixed costs at 80 hm ³ /yr base capacity; scales linearly with installed capacity.
Variable cost (rigid)	€0.20/m ³	Chemicals, membrane replacement, and labour at rigid OEE.
Variable cost (flexible)	€0.14/m ³	30% lower non-energy cost reflecting newer-generation membranes and energy-recovery devices.
Energy intensity	4.5 kWh/m ³	SWRO industry standard [34].
Water tariff (year 0)	€1.30/m ³	Acuamed published wholesale tariff, indexed to energy-price drift.
Discount rate (WACC)	0.08	Reported WACC range for desalination water infrastructure [41].
<i>Operational efficiency</i>		
OEE (rigid)	0.85	Industry-standard for monolithic SWRO plants of Torrevieja vintage [34].
OEE (flexible)	0.90	Reflects newer-generation membranes, ERDs, and staggered-maintenance scheduling.
OEE (commissioning year)	0.65	First-year operating efficiency during start-up and operator familiarisation.
<i>Modular-expansion decision rule</i>		
Module size	5,000 m ³ /day	Standard commercial SWRO train scale.
Maximum modules ($M_{\max}$)	22	Sets the +40 hm ³ /yr expansion envelope at 90% OEE $\times$ 365 days/yr.
Build trigger threshold	0.85	Slightly below the 85–90% range used by [40] to allow for the one-year build lag.
Abandonment threshold	0.70	Reflects structural rather than transient demand decline.
Abandonment window	2 yrs	Distinguishes structural shortfall from single-year low realisation.
Build lag	1 yr	Standard procurement-and-installation horizon for 5,000 m ³ /day modules.
Maximum builds per year	10	Logistical cap consistent with phased deployment of Carboneras and Las Palmas.

9.4 Model Verification and Cross-Validation

The Python Monte Carlo implementation used throughout the thesis is cross-validated against the v7 Excel model under the original parameter inputs (CapEx per module €4.045 M, flexible OEE 90%, build threshold 0.90, demand volatility 10%/yr). Headline outputs agreed within Monte Carlo sampling tolerance, supporting the implementation-level verification claim. The headline analysis in Section 6 uses the updated parameter set (CapEx per module €3.0 M, flexible OEE 90%, build threshold 0.85, demand volatility 15%/yr) reported in Section 4.4 and Table 15.

Table 16: Cross-validation of Python and Excel implementations under identical parameter inputs (original verification calibration).

Metric	v7 Excel	Python implementation
Rigid mean NPV (€M)	512.5	506.1
Flex mean NPV (€M)	592.2	577.3
EVof (€M)	79.7	71.2
P(Flex > Rigid)	0.746	0.717

Differences between the two implementations are within Monte Carlo sampling tolerance for an ensemble of $N = 2,000$ paired trials and do not affect the verdict on flexibility’s value. Convergence of the EVof estimate was verified across $N \in \{250, 500, 1,000, 2,000\}$; the estimate is stable to within $\pm 1\%$ above $N = 1,000$ under common random numbers. The $N = 2,000$ ensemble size is used throughout the thesis to provide additional margin against sampling noise in the sensitivity sub-runs.

9.5 Decision-Support Tool

The Decision-Support Tool is publicly available for evaluation at: [SWRO Decision Tool](#).

The underlying Excel model that the tool reads at startup is available at: [Excel](#).

The tool is implemented in Plotly Dash (Python 3.11), styled with Dash Bootstrap Components, and deployed to Render.com via Gunicorn. Model parameters and calibration data are read from the Excel workbook once at startup through a singleton configuration module; the Monte Carlo engine then runs 2,000 paired trials on demand whenever the user modifies parameters in the Inputs tab. No installation is required.

Native-resolution screenshots of each tab follow.

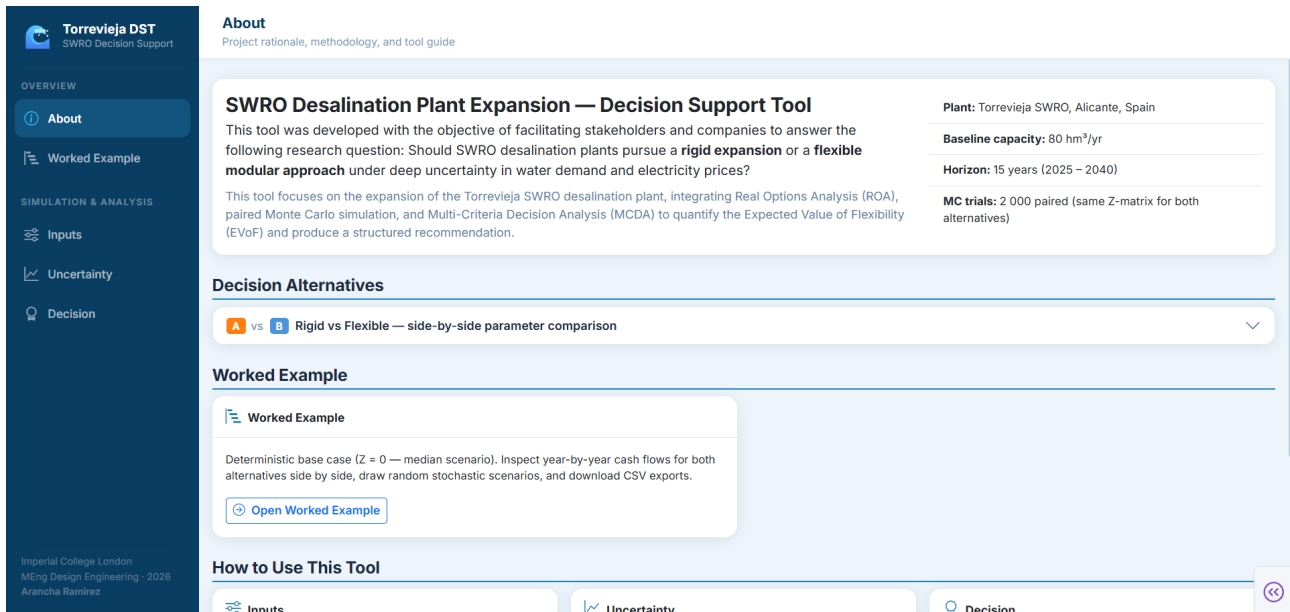


Figure 14: DST About tab.

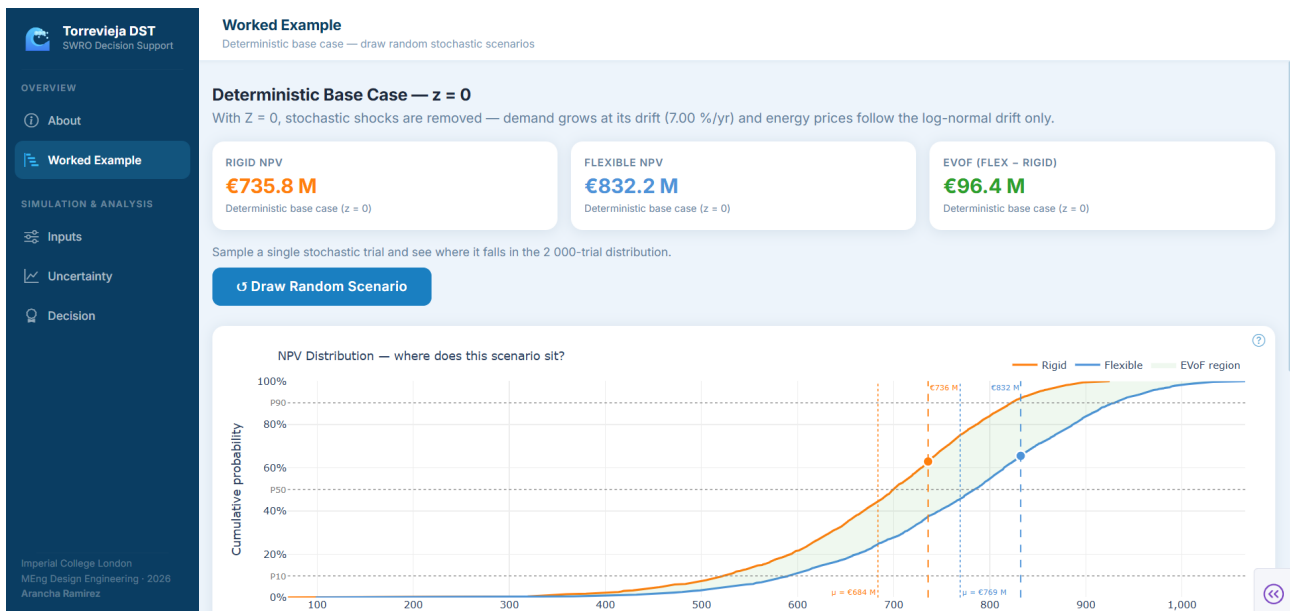


Figure 15: DST Worked Example tab.

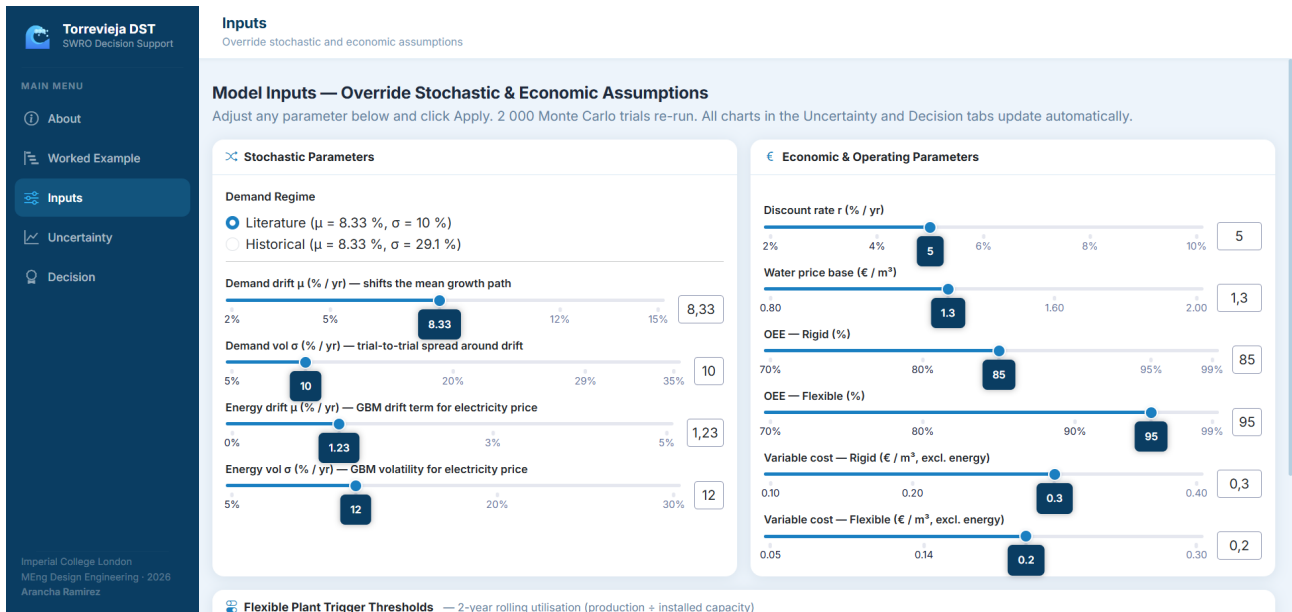


Figure 16: DST Inputs tab.

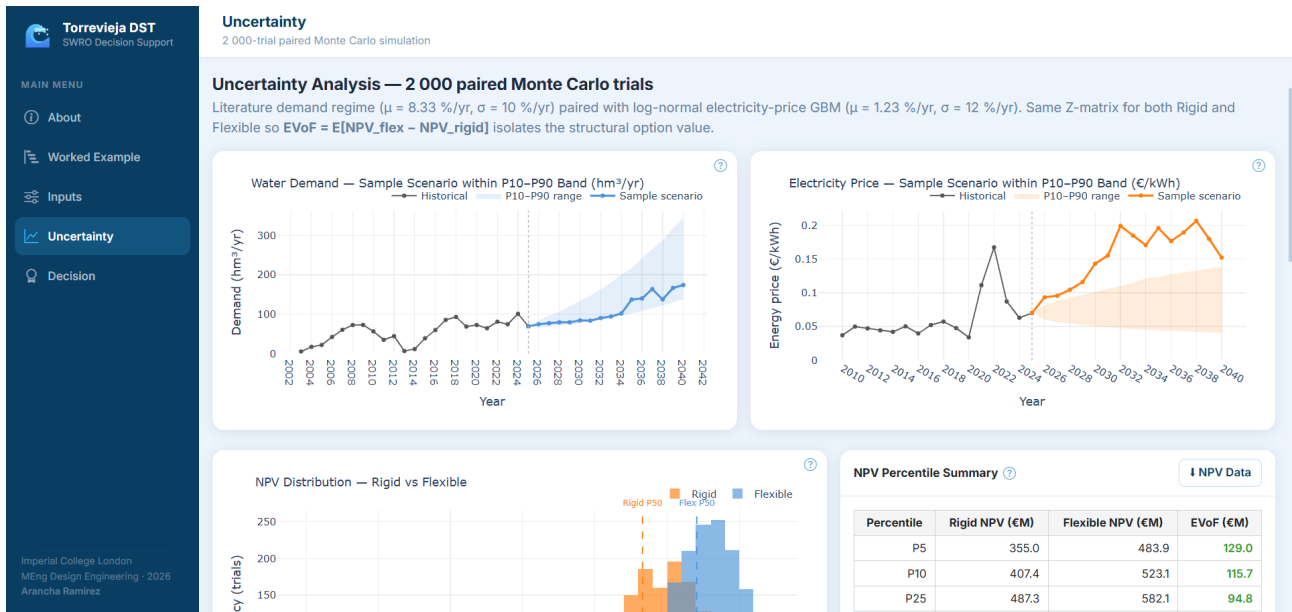


Figure 17: DST Uncertainty tab.

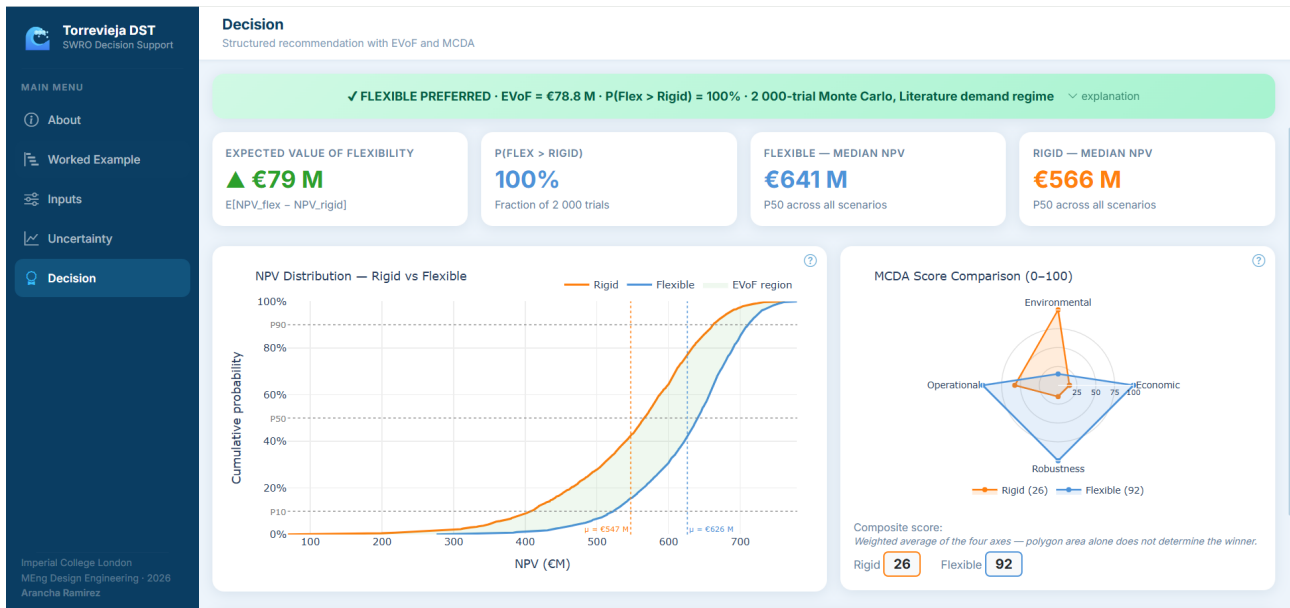


Figure 18: DST Decision tab.